
Block 1: Cognitive Psychology, Learning and Memory

Unit 1: Cognitive Psychology

- **1.0 Introduction**
    - **Cognitive Psychology** is the study of **mental processes** such as perceiving, remembering, and reasoning.
    - Psychologists study mental processes because they believe **behavior is the result of mental processes**.
    - Example: Reading a course book involves perception of words and computation of meaning, driven by goals like completing the course or obtaining a degree.
    - Intentionality, goals, and plans are mental phenomena that affect behavior.
- **1.2 Cognitive Psychology: An Introduction**
    - Cognitive psychology studies **how people perceive, learn, remember, and think about information**.
    - Examples of everyday experiences studied:
        - Overlooking obvious errors when proofreading (pattern recognition).
        - Difficulty taking notes and understanding a lecture simultaneously (attention).
        - Needing to repeat a phone number until dialed (short-term memory).
        - Solving a problem after a break (incubation effect in problem solving).
        - Objects looking farther away on foggy days (perception).
    - Failures of mental processes are noticed and become useful for psychological analysis.
    - Cognitive psychology is the **scientific study of the mind**.
- **1.3 Research Methods in Cognitive Psychology**
    - **1.3.1 Goals of Research**
        - **Data gathering**, **data analysis**, **theory development**, **hypothesis formulation**, **hypothesis testing**, and application to real-world settings.
        - Beyond description, researchers seek to understand **how and why** thinking occurs.
        - Requires inferring beyond direct observation.
    - **1.3.2 Distinctive Research Methods**
        - **i) Experiments on Human Behavior:** Controlled laboratory settings, manipulation of independent variables, measurement of dependent variables, control variables. Allows inference of probable causality.
        - **ii) Psychobiological Research:** Studies the relationship between cognitive performance and cerebral events. Techniques include

post-mortem brain study, brain imaging of individuals with deficits, and imaging normal cognitive activity (e.g., MRI, PET).

- **iii) Self-Reports, Case Studies, and Naturalistic Observation:**
  - **Self-reports:** Individual's own account of cognitive processes.
  - **Case studies:** In-depth studies of individuals.
  - **Naturalistic observation:** Detailed studies of cognitive performance in everyday situations.
  - Useful for hypothesis formulation and describing rare events.
- **iv) Computer Simulations and Artificial Intelligence (AI):** Digital computers influenced cognitive psychology.
  - **Indirect influence:** Models of human cognition based on computer information processing.
  - **Direct influence:** Computer simulations (programming computers to imitate human function) and AI (attempting to create models of the entire human mind). Stimulated discussions on how the human mind functions.

- **1.4 Domains of Cognitive Psychology**

  - **i) Cognitive Neuroscience:** Seeking neurological explanations for cognitive findings and vice versa. Basic electrochemical processes in the brain support cognitive processes from sensation to memory.
  - **ii) Perception:** Detection and interpretation of sensory stimuli. Understanding how we interpret sensory signals. Involves pattern recognition, attention, consciousness, and memory.
  - **iii) Pattern Recognition:** Sensing stimuli as meaningful patterns. Example: Reading involves forming a meaningful pattern from lines and curves to access meaning from memory.
  - **iv) Attention:** Our capacity to process information is limited at sensory and cognitive levels. Overload can lead to breakdown in performance.
  - **v) Consciousness:** "Current awareness of external or internal circumstances". Rejected by behaviorists, but now seriously studied. Example: Awareness of time on a watch leading to a thought about finishing a chapter.
  - **vi) Memory:** Works with perception. Information comes from perception, short-term, and long-term memory (LTM). LTM stores vast numbers of facts and language knowledge, recalled in fleeting seconds.
  - **vii) Representation of Knowledge:** How information is symbolized and combined in the brain. Involves conceptual representation and brain storage/processing.
  - **viii) Imagery:** Mental images of the environment, e.g., cognitive maps.
  - **ix) Language:** Principal means of acquiring and expressing knowledge. Unique abstraction, important component of information processing and storage. Influences perception.
  - **x) Developmental Psychology:** Studies how cognitive structures develop, from childhood to adolescence.

    - **xi) Thinking and Concept Formation:** Thinking is forming new mental representations through information transformation. Ability to think and form concepts is crucial for understanding and processing information.
    - **xii) Human and Artificial Intelligence:** Human intelligence includes acquiring, recalling, using knowledge, understanding concepts, language, following instructions, and rule-based behavior. AI has influenced cognitive science by requiring knowledge of human information processing for program design.
- **1.5 A Brief History of Cognitive Psychology**
    - **1.5.1 Early Thoughts on Thinking:** Fundamental questions: Where does knowledge come from, and how is it represented?.
        - **Empiricists:** Knowledge comes from experience.
        - **Nativists:** Knowledge is based on innate brain characteristics.
        - Ancient philosophers debated the "seat of thought and memory" (e.g., Aristotle: heart; Plato: brain).
    - **1.5.2 Cognition in the Renaissance and Beyond:** Renaissance scholars located knowledge in the brain, acquired through senses and divine sources.
        - **18th Century (British Empiricists):** George Berkeley, David Hume, James Mill, John Stuart Mill suggested internal representations are: (1) direct sensory events, (2) faint copies (memory), (3) transformations of copies (associated thought). These notions form the basis of current research.
        - **19th Century (Early Psychologists):** Gustav Fechner, Hermann Helmholtz, Wilhelm Wundt, William James moved from philosophy to empirical results. Dichotomous theories: structure of mental representation (Wundt, Titchener) vs. processes/acts (Brentano).
        - **William James:** Established first psychological lab in America, wrote "Principles of Psychology" (1890). His view of memory, with both structure and process, links to modern cognitive psychology.
    - **1.5.3 Cognitive Psychology in Early Twentieth Century:** Psychological studies of mental processes became unfashionable, displaced by **behaviorism** for about 50 years. Behaviorists neglected internal states ("intervening variables") focusing on observable behavior.
        - **Edward Tolman (1932):** In "Purposive Behavior in Animals and Men," observed rats learning mazes formed "cognitive maps" of the environment, anticipating modern focus on knowledge representation.
        - **Sir Frederick Bartlett (1932):** In "Remembering," rejected studying memory with nonsense syllables (Ebbinghaus). Argued for meaningful material in naturalistic conditions. Introduced **schema** as a unifying theme for experience, central to modern memory theories.
    - **1.5.4 Cognitive Psychology — As it is Today:** Interest revived in the 1950s for topics like attention, memory, pattern recognition, language, thinking, and consciousness.
        - **Driving forces for resurgence:**

        - **Failure of Behaviorism:** Couldn't account for human behavior diversity (e.g., language). Internal mental processes required investigation.
        - **Emergence of Communication Theory:** Prompted studies in signal detection, attention, cybernetics, information theory.
        - **Modern Linguistics:** New views of language and grammatical structure.
        - **Memory Research:** Verbal learning and semantic organization provided empirical base for memory theories.
        - **Computer Science and AI:** Re-examined problem solving, memory processing/storage, language processing/acquisition. Expanded research capabilities.
        - **Cognitive Development:** Piaget's work on children's concept development with maturation.
    - Increasing evidence suggests internal representations of reality are **not isomorphic** (identical) to external reality; information is stored as **abstract representations** (Tolman, Bartlett) or **neurochemical codes**.
- **1.6 Key Issues in the Study of Cognitive Psychology**

    - These issues can be seen as dialectical (thesis/antithesis) or as a synthesis (both/and).
    - **Nature versus Nurture:** Influence of innate characteristics vs. environment on cognition.
    - **Rationalism versus Empiricism:** Discovering truth through logical reasoning (rationalism) vs. observation and testing through senses (empiricism). Combining theory with empirical methods.
    - **Structures versus Processes:** Should we study the contents/attributes/products of the mind (structures) or the operations of thinking (processes)?.
    - **Domain Generality versus Domain Specificity:** Are cognitive processes limited to single domains or general across many?.
    - **Validity of Causal Inferences versus Ecological Validity:** Should research use controlled experiments for causality or naturalistic techniques for real-world applicability?.
    - **Applied versus Basic Research:** Should research focus on fundamental processes or practical applications?.
    - **Biological versus Behavioral Methods:** Should we study the brain directly (e.g., scanning) or observe behavior (e.g., reaction time)?.
- **1.7 Key Ideas in Cognitive Psychology**

    - **Data** is understood only within an **explanatory theory**, but **theories are empty without empirical data**.
    - Cognition is generally **adaptive**, but not always in specific instances.
    - **Cognitive processes interact** with each other and with noncognitive processes.
    - Cognition needs to be studied through a **variety of scientific methods**.

- **1.7.1 Computer Metaphor and Human Cognition**
    - Early computers excelled at mathematical functions and rule-governed logic, while humans excelled at generalizations, inferences, and understanding complex patterns.
    - **Neural networks** (second generation computers) act more like humans, making generalizations and understanding visual patterns, albeit slower at math.
    - Fundamental differences exist between computer and brain internal workings, but the computer metaphor has had a profound and positive impact.
- **1.7.2 Cognitive Science**
    - A new science created by the convergence of **computer science, neuroscience, and cognitive psychology**.
    - It uses ideas and methods from cognitive psychology, psychobiology, artificial intelligence, philosophy, linguistics, and anthropology.
    - The science of human cognition is being transformed by advances in computer technology and brain science.
- **1.7.3 Neuroscience and Cognitive Psychology**
    - Early cognitive psychology paid little attention to physiological psychology or neuroanatomy.
    - Debates: Is the brain holistic (operations distributed) or localized (activities tied to specific regions)?.
    - **Neurocognitive imagery (e.g., MRI, PET, EEG)** is becoming a major means of investigating cognitive functions, showing both structures and processes.
    - The brain is considered the "engine of cognition".

Unit 2: Information Processing in Learning and Memory

- **2.2 Learning and Memory**
    - **Learning:** Acquiring new knowledge, behaviors, skills, values, preferences, or understanding, involving synthesizing and processing different types of information.
        - Benjamin Bloom (1965) suggested three **domains of learning**:
            1. **Cognitive:** Recall, calculate, discuss, analyze, problem-solve.
            2. **Psychomotor:** Dance, swim, drive.
            3. **Affective:** Like, love, appreciate, fear, hate.
            4. These domains are not mutually exclusive (e.g., learning chess involves all three).
    - **Memory:** The brain's capacity to change in response to experience and retain knowledge. Fundamental for effective actions, understanding, recognizing, decoding, and personal identity.
        - Usually divided into three storage systems: **sensory, short-term, and long-term**.

- **Types of Memory Storage Systems:**
    - **i) Sensory Memory:**
        - Affiliated with **transduction of energy** (converting external energy like light/sound into electrical stimulation for the brain).
        - Very short duration (less than 0.5 second for vision; about 3 seconds for hearing).
        - Retains an **exact copy** of what is seen or heard.
        - **Critical:** Learner must attend to information to transfer it to STM.
        - Information enters STM if it has an **interesting feature** or **activates a known pattern**.
    - **ii) Short-Term Memory (STM) / Working Memory:**
        - Selected information from sensory memory transfers here.
        - Most often stored as **sounds**, but can be images.
        - Functions like a computer's **RAM** – working space for computations, then transfers to LTM or discards.
        - Vulnerable to interruption/interference.
        - Characterized by:
            1. **Limited capacity:** Up to **seven pieces of independent information** (Miller, 1956), known as **Miller's Magic Number (7 ± 2)**. This applies to items with one aspect (e.g., pitch of a tone).
            2. **Brief duration:** 3 to 20 seconds.
            3. **Decay** is the primary mechanism of memory loss.
            4. **Chunking:** Grouping items (e.g., phone numbers into 3-digit and 4-digit groups) increases capacity.
        - **Three basic operations within STM:**
            1. **Iconic memory:** Ability to hold visual images.
            2. **Acoustic memory:** Ability to hold sounds (longer than iconic).
            3. **Working memory:** What we are actively thinking about (conscious memory). Requires active process (e.g., repeating a phone number to dial it).
    - **iii) Long-Term Memory (LTM):**
        - **Relatively permanent storage**.
        - Information stored based on **meaning and importance**.
        - **Transfer from STM to LTM:** Involves **encoding or consolidation** of information, based on **organizing complex information** in STM, not just time spent in STM. Meaningfulness or emotional content plays a greater role.
        - **Chunking** is a proven method for retention in LTM.
        - LTM affects perceptions and influences what we attend to in the environment. Provides framework for new knowledge.
        - **Schemas:** Mental models of the world. Information in LTM is stored in **interrelated networks of schemas**, forming intricate knowledge structures. Activating existing schemas helps process new information.

        - **Top-down processing:** LTM strongly influences perception; prior knowledge affects how we interpret sensory information (e.g., optical illusions).
        - **Rehearsal** is important for retention of learned information in LTM.
- **2.3 Cognitive Information Processing (CIP)**
    - **Information processing:** Change of information detectable by an observer.
    - In cognitive psychology, it's an approach to understanding human thinking, originating in the 1940s-1950s.
    - **Neisser (1967):** Defined cognition as the study of how people **encode, structure, store, retrieve, use, or otherwise learn knowledge**.
    - Hypothesizes intervening variables between environment and behavior, contrasting with behavioral theories.
    - **2.3.1 Principles of the Information Processing System (Huitt, 2000):**
        - **Limited Capacity:** Amount of information processed is constrained (bottlenecks).
        - **Control Mechanism:** Required to oversee encoding, transformation, processing, storage, retrieval, and utilization of information (e.g., executive function). Demands more processing power for new tasks.
        - **Two-way Flow of Information:**
            1. **Bottom-up processing:** Using information from senses.
            2. **Top-down processing:** Using information stored in memory.
            3. Dynamic process for constructing meaning. Analogous to inductive vs. deductive reasoning.
        - **Genetically Prepared:** Humans are predisposed to process and organize information in specific ways (e.g., infants look at human faces, universal language development stages like babbling, cooing).
- **2.4 Information Processing in Learning and Memory**
    - Memory is defined by **brain mechanisms, information type, and operating principles** (Schacter & Tulving, cited in Driscoll, 2001).
    - Memory is a "built store" of connections and representations.
    - Memory as the "general ability... to interpret the perceptual world to help organize responses" (Eliasmith, 2001).
    - **Bartlett's work:** Memory is inaccurate, but systematically so, due to previous information and experiences. Knowledge units are retained, manipulated, and changed.
    - **2.4.1 Cognitive Information Processing Model (CIP) of Learning:**
        - Views learning and memory from a **computer model perspective**.
        - Human mind: takes in information, performs operations (changes form/content), stores, retrieves, generates responses.
        - **Processes: Encoding** (gathering/representing), **storage** (holding), **retrieval** (getting when needed).
        - Guided by a **control process**.

        - Some theorists suggest brain operation resembles parallel processing computers.
        - Highlights **limited capacity of "working memory"** and the concept of **"cognitive load"**.
- **2.4.2 Development of Memory and Information Processing:**
    - **Increased processing capacity, speed, and efficiency** due to maturation and knowledge.
    - **Modifications of neural network connections**.
    - **New emergent concepts** from repeated self-organization.
    - **Increased capacity for problem-solving and metacognition**.
    - **Encoding:** Influenced by maturation (e.g., STM capacity increases with age) and experience (experts remember new info better due to organized knowledge).
    - **Structuring and Organizing:** Ability changes with maturation and experience. Younger children don't rehearse; later learn and spontaneously use strategies (e.g., organizing information, categorizing).
    - **Storage and Retrieval:** Capacity changes over time (e.g., object permanence at 7 months, increasing ability to reproduce action sequences).
    - **Strategy Development (Flavell et al., 2002):**
        - Not linear (can stall or regress).
        - Continues to develop even after mature form is first shown (months/years).
        - Considerable variability in strategy use; children go back and forth.
        - Differing abilities to integrate strategies; ample opportunity for individualized learning programs needed.
- **2.5 Theories of Information Processing**
    - **2.5.1 Designing Instruction:** Educators must ensure new information is processed for LTM retention through **elaboration and connection** to existing memory. Deeper processing and more connections enhance long-term learning.
    - **2.5.2 Bloom's Taxonomy of Cognitive Domain (1956, revised 2000):** Classifies information processing into six levels of complexity.
        - **Knowing:** Repeat what was heard/seen (very little elaboration).
        - **Comprehension:** Summarize/paraphrase (modest elaboration).
        - **Application:** Use concepts/principles to solve problems (more elaboration, significant LTM impact).
        - **Analysis:** Understand relationships among parts and organization into a whole (more elaboration, significant LTM impact).
        - **Synthesis:** Put parts together in an original manner (most complex, highest elaboration).
        - **Evaluation:** Make judgments based on comparison to a standard (most complex, highest elaboration).
        - First four levels are a hierarchy. Synthesis and Evaluation are based on analysis.

- **2.5.3 Sternberg's Information Processing Approach (1988):**
  - Skills-based and continuous development (not staged).
  - Focus on **intelligence** (novice to expert progression).
  - Does not differentiate child/adult learning or incorporate biological development.
  - Intelligence comprised of three kinds of information processing components:
    1. **Metacomponents:** Executive functions; guide planning and decision-making, identify problems, connect to past experiences. No direct action, but direct subsequent actions.
    2. **Performance components:** Actions taken in problem-solving; encode stimuli, hold in STM, make calculations, retrieve from LTM. Weigh merit/consequences of actions.
    3. **Knowledge-acquisition components:** Ability to learn new information to solve potential problems; abstract, may not be directly related to current task.
  - Adds **feedback** and **social interaction** to cognitive development theories (linked to Bruner and Vygotsky). Intelligent behavior defined by sociocultural context, involving adaptation, selection, and shaping of environment.
- Most theories agree: new information is best learned if **matched to existing memory structures** (schemas), which are multi-leveled and have varying specificity.

Unit 3: Neuropsychological Basis of Learning and Memory

- **3.0 Introduction**
  - **Learning:** Process of acquiring new information.
  - **Memory:** Persistence of learning in a state that can be revealed later (Squire, 1987).
  - Learning happens when a memory is created or strengthened, possibly unconsciously through exposure.
  - **Memory stages: Encoding** (acquisition, consolidation), **storage**, and **retrieval**.
    - **Acquisition:** Registers inputs in sensory buffers and analysis stages.
    - **Consolidation:** Creates a stronger representation over time.
    - **Storage:** Creates and maintains a permanent record.
    - **Retrieval:** Utilizes stored information for conscious recollection.
  - Neural basis of memory studied via: a) Case studies of amnesia; b) Animal models; c) Brain imaging.
- **3.2 Memory and Brain**
  - Neuroscientific studies investigate how specific neural circuits enable learning and retention.
  - **3.2.1 Human Memory, Brain Damage and Amnesia:**

        - **Amnesia:** Deficits in memory due to brain damage, disease, or psychological trauma. Can involve inability to learn new things (anterograde), loss of previous knowledge (retrograde), or both.
        - Differential effects on short-term/working memory and long-term memory.
    - **3.2.2 Brain Surgery and Memory Loss:**
        - Early neurosurgical procedures (e.g., temporal lobe resection) accidentally revealed principles of human cognition.
        - **Bilateral removal of the medial temporal lobe, especially the hippocampal formation**, was found to cause severe recent memory loss.
        - **Patient H.M.:**
            - Suffered severe epilepsy; underwent bilateral medial temporal resection in 1953 (age 27).
            - Surgery successfully reduced seizures but caused **profound amnesia**.
            - Could remember life events *prior* to surgery, but **could not form new long-term memories** (severe and permanent inability to acquire and store new information).
            - **Normal short-term memory** (sensory registers and working memory), and intact intelligence, perception, language, personality, motivation.
            - Showed disruption of information transfer from short-term to long-term memory.
        - **Patient R.B.:**
            - Developed dense **anterograde amnesia** similar to H.M. after an ischemic episode.
            - Also had retrograde amnesia extending 1-2 years prior.
            - Brain studies showed damage **restricted to the hippocampus**, specifically CA1 pyramidal cells.
            - This supports the hippocampus's crucial role in **forming new long-term memories** and the distinction between areas that *store* memories and the hippocampus's role in *forming* them.
- **3.3 Memory Consolidation and Hippocampus**
    - **Consolidation:** Process by which memories solidify in long-term stores over days, weeks, months, years. Biologically, it's changes underlying long-term retention.
    - **Hippocampus:** Not the repository of stored knowledge, but critical for the **consolidation of information in long-term memory**.
    - Amnesics' retrograde amnesia (for memories 1-few years prior to damage) supports a consolidation role, not a storage role, for the hippocampus.
    - Consolidation may involve strengthening synaptic connections in the **neocortex**, which then stores the memories (hippocampus no longer required post-consolidation).

    - **Alcoholic Korsakoff's Syndrome:** Amnesia resulting from damage to midline structures of the **diencephalon** (dorsomedial nucleus of thalamus and mammillary bodies), often due to vitamin deficiencies from chronic alcoholism. Causes anterograde and retrograde amnesia.
- **3.4 Anterior and Lateral Temporal Lobes and Memory**
    - Damage to the **lateral cortex of the anterior temporal lobe** can cause dense retrograde amnesia, sometimes extending decades back.
    - This region (including entorhinal and perihippocampal cortex) may be important for **retrieval of information** from distributed representations in the neocortex.
    - **Isolated retrograde amnesia:** Patients with dense retrograde amnesia who can still form new long-term memories, particularly related to anterior temporal lobe damage. This region is not essential for acquiring *new* information.
- **3.5 Animal Models of Memory**
    - Studies in monkeys with medial temporal lobe lesions help understand primate memory systems and human memory/amnesia.
    - **Amygdala:** Data from amnesics (e.g., R.B.) and monkey studies indicate it's **not part of the episodic memory system**, but has a role in **emotion and emotional memories**.
    - Monkey studies (Mishkin, Zola et al.): Lesions producing severe memory deficits involved the **hippocampus AND surrounding cortex** (perirhinal, entorhinal, parahippocampal regions). Lesions restricted *only* to the hippocampus (like R.B.) are severe because the hippocampus needs these vital connections.
    - Animal data is consistent with human amnesics: hippocampal system and associated cortex are critical for **forming long-term memories**. The medial temporal lobe **organizes and consolidates** LTM, which is permanently stored in the neocortex. Short-term memory processes are preserved after medial temporal lobe damage.
- **3.6 Imaging the Human Brain and Memory**
    - Functional brain imaging (PET, fMRI) confirms and extends lesion findings.
    - **3.6.1 Episodic Encoding and Retrieval:**
        - PET studies (Haxby et al., 1996): **Right hippocampus region activated during encoding** of new information (e.g., faces) but not during recognition/retrieval. Consistent with amnesics.
        - **HERA model ("hemispheric encoding-retrieval asymmetry"):** Encoding involves the **left hemisphere** more than the right; retrieval involves the **right hemisphere** more than the left, predominantly dorsolateral prefrontal cortex. Neocortical areas are most activated.
        - Event-related fMRI studies (Wagner et al.): Larger responses in prefrontal and medial temporal regions (parahippocampal cortex) during encoding of items that were later remembered.
    - **3.6.2 Semantic Encoding and Retrieval:**

        - Semantic retrieval involves the **left prefrontal cortex** (including Broca's area). This lateralization holds for objects or words.
        - Evidence for **domain-specific organization** (knowledge of animate/inanimate objects in different regions).
    - **3.6.3 Procedural Memory Encoding and Retrieval:**
        - Amnesics show implicit learning of motor sequences even without explicit memories.
        - PET study (Grafton et al., 1995): Implicit motor learning activated **motor cortex, supplementary motor area (left hemisphere), putamen (basal ganglia bilaterally)**, rostral prefrontal cortex, and parietal cortex. Brain areas controlling limb movements activated.
    - **3.6.4 Perceptual Priming and Implicit and Explicit Memory:**
        - **Perceptual priming (implicit learning):** PET study (Schacter et al., 1996) showed no hippocampus activation, but **decreased blood flow in bilateral occipital cortex** (area 19).
        - Implicit and explicit retrieval are subserved by **separate brain systems**. Deactivation of visual cortex for previously seen words correlates with perceptual priming.
- **3.7 Cellular Bases of Learning and Memory**
    - Memory results from changes in the **strength of synaptic interactions among neurons** in neural networks.
    - **Hebb's law (Donald Hebb, 1949):** If a synapse between two neurons is repeatedly activated at the same time the postsynaptic neuron fires, the synapse will be strengthened. This is **Hebbian learning**.
    - **Long-Term Potentiation (LTP):** Neural circuits in the hippocampus subjected to repeated intense electrical stimulation become more sensitive to stimuli.
        - This enhanced response can last for weeks or longer, suggesting a mechanism for long-term learning.
        - **NMDA receptor:** Doubly gated, located on dendritic spines. Crucial for *producing* LTP (requires excitatory input and postsynaptic depolarization). Chemically blocking NMDA receptors prevents LTP induction.
        - **Non-NMDA receptors:** May be important for *maintaining* LTP.
        - Disrupting LTP (e.g., with drugs or genetic manipulation) impairs learning and remembering (e.g., spatial memory in mice).

Unit 4: Models of Information Processing

- **4.0 Introduction**
    - **Information processing:** The change of information in any manner detectable by an observer.
    - In cognitive psychology, it views cognition as **computational** (mind as software, brain as hardware).
    - Memory research in the 1960s saw formal comprehensive theories emerge.

- **4.2 Waugh and Norman's Model of Primary and Secondary Memory (1965)**
    - **Dualistic theory: Primary Memory (PM)** (short-term storage) is independent of **Secondary Memory (SM)** (longer-term storage).
    - Borrowed from William James's dichotomy.
    - PM has **very limited capacity**; information loss occurs by **displacement** of old items by new ones, not simple decay (once capacity is exhausted).
    - Experiment (16 digits, 1 vs. 4 digits/second): Forgetting rate similar, suggesting **interference is a greater factor than decay** in PM forgetting.
    - PM holds verbal information for verbatim recall (e.g., last part of a sentence). Rehearsal makes it available through SM.
- **4.3 Atkinson and Shiffrin's The Stage Model (1968)**
    - Widely used model, views learning and memory as **discontinuous and multi-staged**.
    - Hypothesizes information is manipulated before storage.
    - Recognizes three stages of memory: **Sensory Memory, Short-Term Memory (STM) / Working Memory, and Long-Term Memory (LTM)**.
        - **Sensory Register:** Holds sensory input briefly (seconds at most). Acts as a portal for all information. **Attention and automaticity** are major influences on transfer to STM.
        - **Short-Term Memory (STM) / Working Memory:** Information attended to and recognized in sensory register passes here. Held for 20-30 seconds. Viewed as **active or conscious memory**.
            - Processed by **rehearsal**:
                - **Maintenance rehearsal:** Repeating over and over.
                - **Elaborate rehearsal:** Linking with other stored information.
            - Capacity: **5 ± 2 units**.
        - **Long-Term Memory (LTM):** Information rehearsed in STM can pass to LTM. Organized into categories, resides for days, months, years, or a lifetime. Retrieval withdraws representation from LTM.
    - **Organizations of Long-Term Memory:**
        - **Categorization (Bruner):** "To perceive is to categorize, to conceptualize is to categorize, to learn is to form categories, to make decisions is to categorize".
        - **Tulving (1972):** Distinguished between **episodic and semantic memory**. Most researchers combine these into **declarative memory**.
            - **Episodic memory:** Sense of remembering actual situations/events; personal experience, specific events.
            - **Semantic memory:** General, abstract information, recalled independently of how it was learned; houses concepts, strategies.
        - **Other organizational types:**

        - **Declarative memory:** Can be talked about or verbalized (conscious thought and sharing), subdivided into semantic and episodic.
        - **Procedural memory:** "How to" knowledge, learned skills, often automaticity (e.g., riding a bike).
        - **Imagery:** Memory structure for pictures, like a photograph.
    - **Information Processing in Three Stage Model:** Atkinson and Shiffrin distinguish **memory** (data retained) from **memory stores** (structural components containing information).
        - Transfer of information from sensory register to STS (short-term store) and STS to LTS (long-term store) is largely **subject-controlled**. Information might enter LTS directly from sensory register.
- **4.4 Level of Recall (Zinchenko, 1962, 1981)**
    - Focus: How a subject interacts with material to be learned affects retention in incidental memory.
    - **Hypothesis:** Words encoded by **deep means** are retained better than those encoded by superficial means.
    - Experiment (word series, different instructions): **Logical connections** (examining meaning) led to greater recall than concrete connections or identifying different meanings.
    - Implied that processing material at different **"levels"** (a learning set) affects recall.
- **4.5 Levels of Processing (LOP): Craik and Lockhart (1972)**
    - Reaction against "boxes-in-the-head" models (like Atkinson-Shiffrin).
    - Memory based on **levels of processing**.
    - Incoming stimuli subjected to series of analyses, from **shallow sensory analysis** to **deeper, more complex, abstract, and semantic analyses**.
    - **Deeper processed items are less likely to be forgotten**.
    - **Levels of analysis:**
        - **Earliest level:** Sensory/featural (e.g., lines and angles of a word).
        - **Deeper level:** Pattern recognition, meaning extraction (e.g., recognizing a letter 'A').
        - **Still deeper level:** Engagement of long-term associations (e.g., word triggers images/stories).
    - **Significant issue:** We can perceive at meaningful levels *before* analyzing at primitive levels. LOP is more a "spread" of processing.
    - **Critiques:** LOP models say little more than that deeply processed items are well remembered, lacking an objective index of depth.
    - **Rehearsal:**
        - **Atkinson-Shiffrin:** Rehearsal transfers information to LTM.

        - **Craik & Lockhart:** Rehearsal either maintains information at one level (maintenance rehearsal, no better retention) or elaborates it to a deeper level (leads to better retention).
    - **Craik and Tulving (1975) Experiment:** Subjects rated words based on structural, phonemic, or semantic aspects.
        - Results: **Deeper processing takes longer** and leads to **better recognition** (semantic aspects > phonological > structural).
- **4.6 Self-Reference Effect (Rogers, Kuiper, & Kirker, 1977)**
    - **Self-reference:** A powerful method variable in levels of processing.
    - Experiment: Subjects rated adjectives on structural, phonemic, semantic, or self-reference tasks.
    - Results: Recall was best for **self-reference words**, then semantic, phonemic, and poorest for structural.
    - Explanation: Self-reference engages a powerful **self-schema** ("I, me, mine"), a rich and elaborate internal network for storing information, leading to better recall due to deep emotional/intellectual investment.
- **4.7 A Connectionist (Parallel Distributed Processing - PDP) Model of Memory: Rumelhart and McClelland**
    - **PDP models:** Neurologically inspired, concerned with human mind's processing mechanisms.
    - Proponents opt for **massively distributed, mutually interactive parallel system** (activities carried out simultaneously through excitation/inhibition of neural cells).
    - Information processing through interactions of **large number of simple processing elements called units** (resemble neurons), sending excitatory/inhibitory signals. Units represent hypotheses (e.g., letters, goals, actions).
    - **Not just a memory model:** Also a model for action and knowledge representation.
    - Units organized into **modules**. Information permeates the model, leaving traces (changes in connection strength/weight) between units.
    - **Memory trace:** Distributed over many different connections.
    - **Content addressable storage:** Information accessed based on its attributes (e.g., recalling a friend's name from picture, address, job). More specific/mutually exclusive cues lead to accurate retrieval.
    - **Interference control:** Inhibitory laws between units prevent interfering components from overrunning the system (e.g., identifying a person as "man" inhibits searches for "women").
    - **Prototype learning (McClelland and Rumelhart, 1986):** Example of a boy seeing different dogs, forming a memory for a "prototypical dog" from combined activation of all exemplar dogs.

- Popular due to elegant mathematical models, relation to neural networks, and flexibility in accounting for diverse memories.

Block 2: Theories and Measurement of Intelligence, Creativity and Problem Solving

Unit 1: Theories of Intelligence (G and S Factors and the Model of JP Das)

- **1.2 The Nature of Intelligence**

    - **Intelligence is hard to define**.
    - Common parlance varies: quickness of answering, leading a successful life.
    - Psychologists also differ.
    - **1.2.1 Theoretical Definitions of Intelligence:**
        - Sum total of knowledge.
        - Ability to learn or profit from experience.
        - Ability to solve problems.
        - Ability to cope with environmental demands.
        - **1921 Symposium:** 13 psychologists had 13 different views. Common elements: (a) **higher-level abilities** (abstract reasoning, problem-solving), (b) **ability to learn**, (c) **adaptation to environment**.
        - **1986 Survey (Sternberg & Determan):** Similar common elements.
        - **Snyderman & Rothman (1987) survey (1000 experts):** Nearly unanimous agreement on **abstract reasoning (99.3%), problem-solving ability (97.7%), and capacity to acquire knowledge (96.0%)** as important elements.
        - **Wechsler (1944, 1975):** "Aggregate or global capacity... to act purposefully, to think rationally and to deal effectively with his environment". "Capacity... to understand the world... and his resourcefulness to cope with its challenges".
        - **Gardner (1983):** "Human intellectual competence must entail a set of skills of problem solving... and must also entail the potential for finding or creating problems".
        - **Sternberg (1997):** "Goal-directed adaptive behaviour".
        - **Common elements in many definitions: adaptation, ability to think abstractly, ability to acquire new information/learn from experience**.
    - **1.2.2 Operational Definition of Intelligence:**
        - **Boring (1923): "Intelligence is what intelligence tests measure**". This sidesteps the conceptual problem but provides a working definition.
        - Most tests assume intelligence is a **general attribute** determining problem-solving ability.

- **1.2.3 Historical Perspective on Individual Differences and Human Abilities:**
    - Systematic study of human abilities began in the late 19th century.
    - **Francis Galton:** Coined "**mental test**," invented first psychological test methods for intelligence/ability. Believed psychological traits inherited. Developed **correlation method**. Credited as "Father of psychological testing and individual differences".
    - **Alfred Binet:** Commissioned by French government to identify mentally deficient children.
        - **Binet-Simon test (1905):** 30 problems to assess child's understanding and reasoning.
        - **1908 revision:** Items arranged by **age levels**, highest level performed successfully was **mental age**.
        - **William Stern (1914):** Suggested dividing mental age by chronological age and multiplying by 100 to get **Intelligence Quotient (IQ)**.
        - Binet's work modeled all subsequent intelligence measurement.
- **1.3 Spearman's Two-Factor Theory of Intelligence (1904)**
    - Charles Spearman used correlational and factor analysis.
    - Critical of Binet for not testing for a general factor.
    - Proposed that intercorrelations between intelligence tests were due to a **general intellective factor ("g")**.
    - Also recognized **specific factors ("s")** unique to particular tests.
    - Believed all intellectual activity contained some "g". "g" was called "**mental energy**".
    - **Tetrad difference:** A mathematical quantity used to test the theory (approximates zero if only "g" and "s" factors are present).
    - Tests with high "g" loadings involve abstract reasoning, comprehension, problem-solving.
    - Later recognized **group factors** (e.g., verbal ability, numerical ability, mental speed, mechanical ability, attention, imagination) that influenced a number of tests beyond "g".
    - **Major contributions:**
        - Developed **mathematical models for "g"** and laid foundation for **factor analysis**.
        - Established a **scholarly tradition** in human abilities investigation.
    - **1.3.1 Critical Appraisal of Two-Factor Theory:**
        - Criticized for being purely psychometric, lacking cognitive theory. Spearman (1923) *did* propose qualitative principles of cognition.
        - Existence of "g" confirmed by Jensen (1998) and Carroll (1993).
        - **1.3.1.1 Thorndike's Theory (1926):**
            - Sharp critic of "g" factor, believed in **many separate characteristics** making up intelligence.

        - Common element resides in the **nature of tasks**, not the individual.
        - Intelligence is a series of **skills or talents**; correlations due to common features in tests.
        - Classified intelligence into (i) social, (ii) concrete, (iii) abstract, but this is task-based, not mental organization.
    - **1.3.1.2 Thomson's Theory (1939):**
        - Intercorrelations result from **common samplings of independent factors**.
        - **Sampling theory:** Every test samples a range of elementary abilities; correlation depends on common units of ability.
        - Believed in a "general ability" like "g" but as a **constant combination of ability elements**, not a basic entity. Group factors are combinations of limited elements.
    - **1.3.1.3 Thurstone's Theory (1935):**
        - Used improved statistical analysis, concluded **seven primary abilities** fit data better than "g".
        - Reasons for conflicting findings with Spearman: population sampled (Thurstone used students), choice of tests (Thurstone used very similar tests).
        - Later work showed **hierarchical structure**, with "g" at the top, primary abilities at a lower level.
- **1.4 Das, Nagliery and Kirby's PASS Theory (1994)**
    - Theory-based, multidimensional view, cognitive in orientation.
    - Based on **Alexander R. Luria's (1966, 1973, 1980) neuropsychological research**.
    - Luria's three primary functional units of cognitive processes:
        - **i) First Unit (Attention-Arousal):** Located in brain stem and reticular activating system. Maintains appropriate cortical arousal and selective attention. Only under optimal arousal can complex attention occur.
        - **ii) Second Unit (Simultaneous and Successive Processing):** Responsible for receiving, analyzing, storing information.
            - **Simultaneous processing:** Associated with occipital-parietal areas. Elements related to every other element (surveyability). Example: Comprehending relationships among shapes in "draw a triangle above a square...".
            - **Successive processing:** Associated with fronto-temporal areas. Integrates stimuli into a specific serial order, each component related to the next. Example: Decoding/producing syntax in language.
        - **iii) Third Unit (Planning):** Located in prefrontal divisions of frontal lobes. Responsible for programming, regulation, and control of mental activity

(executive functioning). Involved in decision-making, problem-solving, self-monitoring, voluntary activity, impulse control, linguistic skills.

- **Knowledge Base:** Integral component, represents all information from cultural and social background, determines mental activity form.
- **Output/Action and Behavior:** Final component.
- Cognitive Assessment System (CAS) developed to measure proficiency in PASS processes.
- **1.4.1 Critical Appraisal of PASS Theory:**
  - Novel approach to intelligence assessment, cognitive and neuropsychologically grounded.
  - Moves away from conventional tests, provides theory-based multidimensional view.
  - Practical utility: CAS allows examining relative contribution of cognitive processes in testing.

Unit 2: Multiple Theories of Intelligence (Guilford, Gardner and Sternberg)

- **2.2 Guilford's Structure-of-Intellect (SOI) Theory**

  - Intelligence as **combination of multiple, relatively independent abilities**, not a single entity.
  - Applied factor analytic method.
  - Rejected "g" factor and broad factor groups like Thurstone's.
  - **SOI model (1956, modified 1977):** Defines intelligence as a systematic collection of abilities for processing different information kinds in various ways.
  - Represented as a cube with three dimensions:
    - **Contents Dimension (5 types):** Broad areas of information.
      - **Visual:** Information from retinal stimulation (image).
      - **Auditory:** Information from cochlear stimulation (image).
      - **Symbolic:** Symbols/signs with no inherent meaning (e.g., numbers, alphabet letters).
      - **Semantic:** Words/sentences (oral, written, mental).
      - **Behavioral:** Acts of individuals.
    - **Operations Dimension (5 types):** General intellectual processes.
      - **Cognition:** Understand, comprehend, discover, become aware.
      - **Memory:** Memorize information.
      - **Divergent Production:** Generate multiple solutions.
      - **Convergent Production:** Deduce a single solution.
      - **Evaluation:** Judge ideas/solutions.
    - **Products Dimension (6 types):** Results of applying operations to contents.
      - **Unit:** Single item of information.
      - **Class:** Set of items sharing attributes.
      - **Relation:** Connection between items/variables (opposites, associations, sequences, analogies).

        - **System:** Organization of items/networks with interacting parts.
        - **Transformation:** Changes in perspectives, conversions, mutations to knowledge (e.g., reversing letters).
        - **Implication:** Predictions, inferences, consequences, anticipations.
    - Original: 4 Contents x 5 Operations x 6 Products = 120 factors.
    - Modified (1977): 5 Contents (Visual/Auditory split) x 5 Operations x 6 Products = **150 cells/factors**. Over 100 empirically verified.
    - Example: Studying unrelated words to recall (Semantic content, Memory operation, Units product).
    - **2.2.4 Evaluation:**
        - More complex than Thurstone's.
        - Criticized for narrow, specialized factors with little predictive value (e.g., educational/vocational guidance).
        - Concentrated studies on restricted intelligence ranges, reducing "g" scope.
        - Nevertheless, significant impact on intelligence research.
- **2.3 Gardner's Theory of Multiple Intelligences (1983)**
    - Claimed most previous conceptualizations of intelligence were too narrow.
    - Intelligence is a **bio-psychological potential to process information** activated in a cultural setting to solve problems or create valuable products.
    - Not a single entity, but **many intelligences**.
    - Established **eight criteria** for defining "multiple intelligences" (beyond psychometric results):
        - Brain isolation by damage (potential).
        - Place in evolutionary history.
        - Presence of core operations.
        - Susceptibility to encoding.
        - Distinct developmental progression.
        - Existence of idiot-savants, prodigies, exceptional people.
        - Support from experimental psychology.
        - Support from psychometric findings.
    - **Initially 7 intelligences, added Naturalistic (8th) in 1999, considered Existential (9th)**.
    - **The Nine Intelligences:**
        - **Linguistic Intelligence:** Effective use of language for expression, rhetoric, poetry, remembering information. (Writers, poets, lawyers, speakers).
        - **Logical-mathematical Intelligence:** Analyze problems logically, carry out mathematical operations, investigate scientifically. (Scientists, physicists, mathematicians, engineers).
        - **Musical Intelligence:** Skill in performance, composition, appreciation of musical patterns, recognizing pitches, tones, rhythms. (Instrumentalists, singers, composers).

        - **Bodily-kinesthetic Intelligence:** Expertise in using one's whole body or parts for expression/problem-solving. (Dancers, athletes, surgeons).
        - **Spatial Intelligence:** Recognizing/manipulating patterns of wide spaces (pilots, navigators) and confined spaces (sculptors, architects, chess players).
        - **Interpersonal Intelligence:** Understand intentions, feelings, motivations of others. (Sales, teaching, counseling, politics).
        - **Intrapersonal Intelligence:** Understand oneself, appreciate feelings, fears, motivations; effective working model of self. (Philosophers, psychologists, theologians).
        - **Naturalistic Intelligence:** Related to nature, nurturing, classifying flora/fauna. (Naturalists, farmers, gardeners).
        - **Existential Intelligence:** Contemplate phenomena beyond sensory data (infinite, infinitesimal). (Cosmologists, philosophers).
    - Intelligences rarely operate independently; used simultaneously and complement each other.
    - Critiqued traditional schooling's focus on linguistic and logical-mathematical intelligence, advocating for diverse teaching methods.
    - **2.3.10 Evaluation:**
        - Profound impact on education.
        - Useful for investigating specialized functions or mental dysfunctions.
        - **Major objection:** Excessive importance on independence of intelligences; brain acts as a whole.
        - **Criticisms:**
            - Intelligences are cognitive **styles** rather than stand-alone constructs.
            - Not empirical enough to generate measurement models.
            - "Different abilities" rather than "different intelligences" (Eysenck).
            - Unintelligent to describe someone tone-deaf or uncoordinated as unintelligent.
            - Incompatible with "g" factor, heritability, environmental influences, and broadens intelligence meaninglessly.
        - Still consistent with investigations of distinct talents and skills.
- **2.4 Sternberg's Triarchic Theory of Intelligence (1985)**
    - Believed previous theories incomplete.
    - **Triarchic theory:** Includes three facets or subtheories that interact.
        - **Componential (Analytical) Subtheory:** Outlines structures/mechanisms underlying intelligent behavior. Relates to individual's internal world. Similar to standard psychometric intelligence.
            - **Metacomponents:** Control, monitor, evaluate cognitive processing; executive functions for planning and strategy. Decide *what* to do.

            - **Performance Components:** Execute strategies; basic operations (encode stimuli, STM, calculations, retrieve from LTM). Actually *do* it.
            - **Knowledge Acquisition Components:** Processes for gaining and storing new knowledge (capacity for learning).
        - **Experiential (Creative) Subtheory:** Interprets intelligent behavior along a continuum from novel to familiar tasks/situations. Ability to successfully deal with new/unusual situations by drawing on existing knowledge. Involves insights, synthesis, reacting to novel stimuli.
            - Splits experience into **novelty** and **automation**.
                - **Novelty:** Adept at managing new situations, finding new solutions.
                - **Automation:** Performing processes multiple times with little thought, allowing parallel processing.
        - **Contextual (Practical) Subtheory:** Specifies intelligent behavior defined by socio-cultural context. "**Street smarts**".
            - Processes:
                - **Adaptation:** Change oneself to adjust to surroundings (e.g., wearing warm clothes).
                - **Shaping:** Change environment to suit needs (e.g., teacher setting new rule).
                - **Selection:** Find a completely new, alternate environment (e.g., immigrants seeking better life).
    - Intelligent behavior arises from a **balance** between analytical, creative, and practical abilities, functioning collectively for success in socio-cultural contexts.
    - Central feature: **adaptability** (within individual and context).
    - **2.4.4 Evaluation:**
        - Useful for understanding human intelligence, captures aspects beyond conventional theories.
        - Differs from Gardner's (emphasizes processes vs. domains) and emotional intelligence.
        - **Criticisms:**
            - **Unempirical nature**.
            - Claim that traditional IQ tests don't measure practical intelligence is challenged by correlation with income.
            - Practical intelligence seen as task-specific skills, not broad cognition.
            - Difficulty translating to psychometric instruments.
            - Question whether creativity is a separate cognitive ability or a product of high general intelligence.

Unit 3: Measurement of Intelligence

- **3.0 Introduction**

- Intelligence tests evaluate **general level of cognitive functions and intellectual ability**.
- Used for academic classification, occupational counseling, personnel selection.

- **3.2 Theoretical Background of Measurement of Intelligence**
  - Test content and measurement vary based on theoretical background.
  - **3.2.1 General Factor Theories:**
    - Claim intelligence is a **general intellectual capacity (g-factor)** common to all activities.
    - **Charles Spearman (1904):** Proposed "g" factor and specific "s" factors. High positive correlation between tests indicates shared "g". Distinct "s" factors should have zero correlation.
    - **Jensen (1998):** Argued "g" is manifested in many ways, testable by unlimited variety of items. Controversial view of "g" being largely inherited and rarely influenced by environment.
    - **Binet and Simon (1916):** Conceptualized intelligence as a general intellectual capacity measured by variety of discrete abilities. Integrated score (IQ) is true measure. "g" includes abstract reasoning, comprehension, clear thought direction, purposeful thinking, self-corrective judgment.
  - **3.2.2 Multi-factor Theories:**
    - Neglect "g" factor, describe intelligence based on **groups of traits or factors**.
    - **3.2.2.1 Thurstone's Primary Mental Abilities/Group Factor Theory:**
      - Certain mental operations have a "primary" factor, giving psychological/functional unity, differentiating from others.
      - **Seven primary factors:**
        - **Number Factor (N):** Rapid and accurate numerical calculations.
        - **Verbal Factor (V):** Verbal comprehension.
        - **Space Factor (S):** Manipulating imaginary objects in space.
        - **Memory (M):** Memorize quickly.
        - **Word Fluency (W):** Think of isolated words rapidly.
        - **Reasoning (R):** Discover rule/principle in series/groups.
        - **Perceptual Speed (P):** Note visual details rapidly.
      - Led to "Test of Primary Mental Abilities (PMA)".
    - **3.2.2.2 Guilford's Model of Structure of Intellect (1967):**
      - Three-dimensional model (Content, Operation, Product).
      - (Details covered in Unit 2, Block 2)
    - **3.2.2.3 Gardner's Theory of Multiple Intelligences (1983):**
      - Humans have potential for multiple intelligences, overlooked by traditional testing.

        - Eight (or nine) intelligences including linguistic, logical-mathematical, musical, bodily-kinesthetic, spatial, interpersonal, intrapersonal, naturalistic (and existential).
        - (Details covered in Unit 2, Block 2)
    - **3.2.2.4 Sternberg's Triarchic Theory (1985):**
        - Three types of intelligence: (i) Analytical, (ii) Creative, (iii) Practical.
        - (Details covered in Unit 2, Block 2)
    - **3.2.2.5 Vernon's Hierarchical Theory:**
        - Fills gaps between Spearman's and Thurstone's theories.
        - Intelligence comprises abilities at varying levels of generality:
            - **Highest level:** "g" (general intelligence).
            - **Next level:** Major group factors (e.g., verbal-numerical-educational (v.ed), practical-mechanical-spatial-physical (k.m.)).
            - **Next level:** Minor group factors.
            - **Bottom level:** "s" (specific) factors.
        - Studied genetic and environmental factors: Individual differences ~60% attributable to genetic factors.
- **3.3 History of Measurement of Intelligence**
    - **3.3.1 Galton and Cattell:**
        - **Sir Francis Galton:** First institutional effort to measure individual differences. Believed intelligence measurable by **sensory discrimination**.
        - **James McKeen Cattell (1890):** Coined "mental test," described tests measuring muscular strength, reaction time, memory, etc..
    - **3.3.2 Contribution of Alfred Binet:**
        - Developed tasks to measure complex mental facilities (memory, attention, comprehension) for identifying mentally retarded children.
        - **Binet-Simon scale (1905):** 30 pass/fail items, combination of mental/physical strategies.
        - **Breakthrough:** Complexity of tasks, breadth of abilities, measurable in clinical interview (not just labs).
        - Innovations: strict administration, age-graded norms, rank order of items, first estimate of **mental age**.
        - **1908 revision:** Items arranged by age levels.
        - **1911 revision:** Extended age range to adulthood.
        - **Stanford-Binet Intelligence Scale (1916):** Terman's revision, comprehensive, rigorous standardization, extensive manual.
    - **3.3.3 The Concept of IQ:**
        - **William Stern (1912):** Proposed dividing mental age by chronological age and multiplying by 100 to get IQ. Terman incorporated this into Stanford-Binet 1916.

- **3.3.4 World War I and Army Personnel Selection:**
  - **1917:** Committee led by Robert M. Yerkes prescribed intelligence tests for rapid classification.
  - **Army Alpha:** Language test.
  - **Army Beta:** Non-language performance test.

- **3.4 Major Intelligence Tests**
  - **Classifications:**
    - **Individual vs. Group tests:** Administered to one person vs. multiple simultaneously.
    - **Verbal vs. Non-verbal tests:** Use written language (literacy required) vs. manipulative items (for illiterates, children, deaf, different cultures).
  - **3.4.1 Types of Intelligence Scales:**
    - **3.4.1.1 Stanford-Binet Test:**
      - **Revisions:** Terman/Merrill (1937, L & M forms), Merrill (1960), SB4 (1986), SB5 (latest, 2003).
      - **SB4:** Hierarchical model, assessed verbal, abstract/visual, quantitative reasoning, short-term memory. Provided overall IQ and composite scores.
      - **SB5:** Adheres to **Cattell-Horn-Carroll (CHC) cognitive theory**. Age range 2-89 years, individually administered, 10 subscales.
      - **Assessed areas:** General cognitive functioning, verbal/nonverbal intelligence, five CHC factors. Nonverbal portion is 50% of test.
      - **Five CHC factors measured: Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing, Working Memory**.
      - **Scoring:** 10 subtest scaled scores (mean 10, SD 3) combine into factor index, domain (Nonverbal IQ, Verbal IQ), abbreviated, and Full Scale IQ (mean 100, SD 15).
      - **Change-Sensitive Scores (CSSs):** Use item response theory to compare score changes over time. Useful for extreme performance levels.
    - **3.4.1.2 Wechsler Intelligence Scales:**
      - First scale in 1939. Successive revisions for adults (WAIS), school children (WISC), preschoolers (WPPSI).
      - **Wechsler (1944):** Defined intelligence as global capacity to act purposefully, think rationally, deal effectively with environment, but also as an aggregate of specific, qualitatively different abilities.
      - **3.4.2.1 Structure of WAIS-IV (2008):** 10 core subtests (comprise Full Scale IQ) and 5 supplemental.
      - **Four index scores (major components):**
        - **Verbal Comprehension Index (VCI)**.
        - **Perceptual Reasoning Index (PRI)**.
        - **Working Memory Index (WMI)**.

        - **Processing Speed Index (PSI)**.
    - **Two broad scores:**
        - **Full Scale IQ (FSIQ):** Total combined VCI, PRI, WMI, PSI.
        - **General Ability Index (GAI):** Based only on VCI and PRI subtests (less vulnerable to impairment).
    - **3.4.2.2 Subtests (Examples):**
        - VCI: Similarities, Vocabulary, Information, Comprehension (supplemental).
        - PRI: Block Design, Matrix Reasoning, Visual Puzzles, Picture Completion (supp.), Figure Weights (supp.).
        - WMI: Digit Span, Arithmetic, Letter-Number Sequencing (supp.).
    - **3.4.2.3 PSI (Processing Speed Index) Includes:** Symbol Search, Coding, Cancellation (supp.).
    - **Interpretation:** Uses **deviation IQ score** (standard scores, mean 100, SD 15) for meaningful peer comparison. WAIS-IV standardized on 2,200 US individuals (16-90 yrs).
- **3.4.3 Kaufman Assessment Scales:**
    - **Kaufman Assessment Battery for Children (K-ABC; 1983):** Bridged gap between intelligence theories and measures.
    - **3.4.3.1 KABC-II (2004):** Age 3-18 years. Measures **learning (LTM retrieval), sequential processing (STM), simultaneous processing (visualization), planning (fluid ability), verbal knowledge (crystallized ability)**.
    - **Theoretical Models:** Founded on **Luria's (1973) neuropsychological model** (three blocks) and **Cattell-Horn-Carroll (CHC) approach**.
    - **Global Scores:**
        - **Mental Processing Index (MPI):** From Luria perspective, excludes acquired knowledge.
        - **Fluid-Crystallised Index (FCI):** From CHC perspective, includes acquired knowledge.
    - Examiners choose interpretive system.
    - **3.4.3.2 Structure of KABC-II (Subtests categorized by Scales):** Sequential/Gsm, Simultaneous/Gv, Planning/Gf, Learning/Glr, Knowledge/Gc.
    - **Standard Scores:** MPI and FCI have mean 100, SD 15. Subtests have mean 10, SD 3.
    - **3.5.1 Kaufman Adolescent and Adult Intelligence Test (KAIT, 1993):**
        - Individually administered for 11-85+ years.
        - Integrates Horn & Cattell's fluid/crystallized intelligence, Luria/Golden's frontal lobe planning, Piaget's formal operational thought.

        - **Crystallized scale:** Measures concepts from schooling/acculturation.
        - **Fluid scale:** Measures ability to solve new problems.

Unit 4: Creativity and Problem Solving

- **4.2 Creativity**
    - **Definition:** Goal-directed thinking that is **unusual, novel, and useful**.
    - Differs from intelligence; complex mix of motivational, personality, environmental, chance factors, and products.
    - **Historical Creativity:** Ideas that influence entire human civilization (e.g., Mona Lisa, laws of thermodynamics).
    - **Newell, Shaw, and Simon (1963) criteria for creativity:**
        2. Novelty and usefulness.
        3. Rejects previously accepted ideas.
        4. Requires intense motivation and persistence.
        5. Results from organizing an unclear situation in a coherent, clear, and new way.
    - **Sternberg (2006) commonalities in creativity research:**
        2. Produces novel and compelling ideas/products.
        3. Has domain-specific and domain-general elements.
        4. Measurable to some extent.
        5. Can be developed and promoted.
        6. Not highly rewarded in practice as in theory.
    - **Runco (2007):** "Products definitions" vs. focusing on the **process** of creative thinking.
    - **Process characteristics:** Imaginative, purposeful, produces something original (relative to self, peers, field), has value. Involves idea generation AND evaluation.
    - **Beghetto and Kaufman (2007) types of creativity:**
        2. **"little-c" (everyday creativity):** Novel and personally meaningful interpretation of experiences, actions, events (intrapersonal judgment).
        3. **"Big-C" (eminent creativity):** Relies on interpersonal and historical judgments of novelty, appropriateness, and lasting impact.
- **4.2.2 Investment and Confluence Theory of Creativity (Sternberg, 2006)**
    - Creativity requires a **confluence of six distinct but interrelated resources**:
        2. **Intellectual skills:**
            - **Synthetic skill:** See problems in new ways, escape conventional thinking.
            - **Analytic skill:** Recognize which ideas are worth pursuing.
            - **Practical-contextual skill:** Persuade others of idea's value.
            - Confluence of all three is important.
        3. **Knowledge:** Enough knowledge to move a field forward, but not so much that it constrains thinking. Can help or hinder.

4. **Thinking styles:** Preferred ways of using skills (decisions on how to deploy skills). **Legislative style** (preference for thinking in new ways) is important. Ability to think globally as well as locally.
5. **Personality:** Traits like willingness to take risks, perseverance, openness to new experiences (Note: Source mentions this in a later section regarding impediments, but it's relevant here implicitly).
6. **Motivation: Intrinsic, task-focused motivation** is essential (love for the work, not just rewards). One decides to be motivated.
7. **Environment:** Supportive and rewarding of creative ideas (e.g., forum for proposing ideas).

- **Confluence:** Creativity is more than simple sum; includes thresholds (e.g., knowledge below which creativity isn't possible), partial compensation (strength in one component counteracts weakness in another), and interactions (high levels multiplicatively enhance).

- **4.2.3 Aspects of Creativity (Torrance Tests of Creative Thinking - TTCT, Torrance 1966, 1974)**

  - Guilford (1986) linked creative thinking to **divergent5. **Abstractness of Titles:** Degree beyond labeling; abstraction of thought. 6. **Resistance to Premature Closure:** Degree of psychological openness; considering variety of information.
  - (Note: Flexibility scale deleted in 1990 due to high correlation with fluency).

- **4.2.4 Stages of Creativity (Graham Wallas, 1926)**

  - Creative thinking follows four successive steps:
    2. **Preparation:** Gather information about the problem, attempt solutions (trial-and-error). Learn as much as possible, list concepts, don't worry about redundancy.
    3. **Incubation:** Solution exists but is unclear. Subject does not intentionally work on problem, allows it to sink into unconscious. Relax and reflect.
    4. **Illumination (Insight):** Sudden insight into the problem; new solution, idea, or relationship emerges. Reformulate or formulate new ideas; more conscious work.
    5. **Verification:** Test, try, and check the solution created. May modify ideas. Rework structure to represent interrelationships.
  - Stages may appear in different order, combine, or not occur regularly (e.g., knowledge may skip preparation).

- **4.2.5 Creativity and Intelligence**

  - **Extreme opinion:** Creativity and intelligence are totally independent; intelligence does not influence creativity. Depends on domain-specific knowledge and deliberate practice.
  - **Terman's (1925) study:** High IQ insufficient for creativity. Most highly gifted children (mean IQ 151) achieved occupational success, but none showed noteworthy creativity.

- **Weak relationship:** Torrance (1977) reported median correlation between IQ and TTCT was only **.20**. Factor analyses yielded separate factors.
- **Threshold hypothesis (Guilford, 1967):** A **minimal level of IQ (arbitrarily set to 120)** is necessary but not sufficient for creativity. Creative achievement impossible below this threshold. Scatter plots show triangular pattern (no low IQ/high creativity).
- **Certification hypothesis (Hayes, 1989):** Doubts intrinsic links; suggests creative occupations (e.g., architecture, science) simply require high formal education, which correlates with IQ. Society denies creative outlets to those without high academic performance.

- **4.2.6 Measurement of Creativity**

  - Most creativity tests measure **divergent thinking**.
  - **Torrance Test of Creative Thinking (TTCT):** Most widely used, extensive reliability/validity research (Kim, 2006). Translated into over 30 languages. Based on Guilford's SOI battery.
  - **Two forms:** TTCT-Verbal and Figural (with parallel tests A and B).
  - **Measures (since 1990):**
    2. **Fluency:** Total number of relevant responses.
    3. **Originality:** Rarity of ideas (statistically infrequent).
    4. **Elaboration:** Number of added ideas.
    5. **Abstractness of Titles:** Degree beyond concrete labeling.
    6. **Resistance to Premature Closure:** Degree of psychological openness.
  - **Criticisms against TTCT:**
    2. Response set might influence results.
    3. Different conditions lead to performance differences.
    4. Rater inconsistency.
    5. Inadequate test structure.

- **4.3 Problem Solving**

  - **Definition:** Thought process involved in a person's effort to **remove obstacles** to achieve a goal state. Involves critical thinking, decision-making, solving routine problems.
  - **4.3.1 Terms Related to Problem Solving:**
    2. **Original/Initial State:** Situation at the beginning of the problem.
    3. **Goal/End State:** Desired outcome.
    4. **Obstacles:** Things blocking goal attainment.
    5. **Person/Operator:** Individual striving to reach goal, manipulating elements internally with symbols/images.
    6. **Problem Space:** Initial state, goal state, and operators combined. Includes elements to be organized. Requires coherence, correspondence to background knowledge.
    7. **Rules:** Exist for converting problem state to goal state.

- **Example (Kohler's Chimpanzee):** Chimpanzee (operator) in cage, bananas (goal state), boxes (elements) to reach bananas (rule: stack boxes). Sudden solution termed **insight**.

- **4.3.2 Typologies of Problems:**

  - **Well-defined problems:** Definable initial/goal states, definite operators, clear/explicit rules (e.g., mathematical equations, jigsaw puzzles).
  - **Ill-defined problems:** One or all elements of problem space (initial state, goal state, operators, rules) are unclear (e.g., creating a painting).
  - **Greeno (1978) classification of well-defined problems:**
    2. **Problems of Inducing Structure:** Identify existing relationships among elements, construct new relationship (e.g., analogies like "bird to sky as fish is to water"). Requires attribute discovery, encoding, comparing encoded attributes.
    3. **Problems of Transformation:** Find sequence of operations to transform initial state to goal state (e.g., Tower of Hanoi). Primarily requires **planning** using **means-end analysis** (identifying differences and reducing them).
    4. **Problems of Arrangement:** Rearrange/recombine elements to satisfy criteria (e.g., anagrams, jigsaw puzzles). Cognitive skill needed: **constructive search**.
- **4.3.3 Stages of Problem Solving:**

  - **Gestalt psychologists:** Similar to creative thinking: **preparation, incubation, illumination/insight, verification**.
  - **Polya's Four Stages:**
    2. **Define, understand, think about the problem:** Identify problem, attributes, knowledge area, collect info.
    3. **Devise a plan for solution:** Think of alternate ways, prepare flowchart.
    4. **Carry out the plan:** Execute the solution.
    5. **Looking back:** Verify solved problem, check reasonableness, constraints, communicate results.
  - **Information Processing Translation of Polya's Stages:** Encoding in working memory, search LTM for plan, execute production system, evaluate results.
- **4.3.4 Strategies of Problem Solving:**

  - **Strategy:** Set of sequential steps/procedure used to arrive at a solution.
  - **i) Algorithms:**
    2. Strategy that **ensures correct solution** if well-defined rule is followed properly.
    3. Properties: Exact steps, terminates, effective (correct answer), general (solves every instance).
    4. Example: Systematic random search for anagrams (trying all possible letter sequences).

5. Rarely used by humans due to high effort/time.

- **ii) Heuristics:**
  2. General suggestions or "**rules of thumb**" useful for various problems.
  3. **Do not ensure a correct solution**. Powerful and general.
  4. General heuristics are context-free; specific heuristics for specialized areas.
  5. **Types of Heuristics:**
     - **Means-end analysis:** Most common/general. Divide problem into sub-problems, detect difference between current/goal state, reduce difference. Example: Algebraic equation solving.
     - **Working backward:** Start at goal state, move backward to initial state. Useful when end state is uniquely well-defined (e.g., maze).
     - **Planning strategy/Analogy:** Divide problem into simple/complex aspects; solve simple first. Use solution from an earlier problem to help with current one.
     - **Thinking Aloud (Ericsson & Simon):** Concurrent verbalization of thoughts while solving problem. Provides "protocol" (description of solution process). Can sometimes interfere with non-reportable/unconscious processes leading to insight.
     - **Generate and Test:** Generate possible solutions, then test them (e.g., finding words starting with 'c' that are eatables). Loses effectiveness with many possibilities.
     - **Lateral thinking:** Indirect, creative solutions.
     - **Method of focal objects:** Synthesizing non-matching characteristics.
     - **Reduction:** Transforming problem into another with existing solutions.
     - **Research:** Employing existing ideas or adapting solutions.
     - **Root cause analysis:** Eliminating problem cause.
     - **Trial-and-error:** Testing possible, difficulty level. Larger problems are harder and more time-consuming.
  6. **Degree of difference between initial and goal state:** Greater difference decreases likelihood of solution; more disorganized problem space.
  7. **The perceiver's Set (Mental Set):** Tendency to perceive and respond to a stimulus in a stereotypical manner based on prior experience. Can be facilitatory or inhibitory.
     - Example: Pronouncing "MACHINERY" as "MacHinery" due to prior "MACDONALD".
     - Minimized by increasing time interval, explicit instructions, exceptions.
  8. **Functional Fixedness:** Tendency to perceive objects with their **customary and stereotypical use**, hindering ability to see novel uses.
     - Example: Duncker's Candle Problem (seeing box only as container, not platform).

        - Example: Simmel's Coin Problem (dividing 8 coins into 4+4 vs. 3+3+2).
    9. **Lack of Problem Specific Knowledge or Expertise:** Experts vs. novices approach problems differently.
        - **Experts:** Organize problems at deeper, principled level; spend more time qualitatively analyzing; check for errors.
        - **Novices:** Represent info superficially, focus on objects, plunge into solutions.
        - Example: Chess masters extract more info from brief exposures if pieces configured as a game.
        - Expertise alone is not always enough; medical/mental status can affect problem solving.

Here are comprehensive notes for your exam, arranged by block, unit, and topic, drawing directly from the provided sources. Key concepts are bolded for clarity, and bullet points are used for lists and better readability.

Block 3: Language and its Cognitive Processes

Unit 1: Language Acquisition

- **1.0 Introduction**
    - **Language** is a system of communication where thoughts are transmitted via sounds (speech, music) or symbols (written words, gestures).
    - Cognitive psychology has been interested in language and thought since the 1950s.
- **1.2 Language and Cognition – Why Study Language?**
    - Human language development shows unique **abstraction**, foundational to cognition.
    - Language processing is a key part of **information processing and storage**.
    - Human **thinking and problem solving** often involve language (e.g., abstracting puzzles into verbal symbols).
    - It is the **main means of human communication**.
    - Language **influences perception**, and perception influences language, showing an interdependency.
    - Specific **cerebral areas** (neuroanatomical structures) are engaged in processing words, speech, and semantics.

- **1.3 Linguistics**
    - **Linguistics** is the formal description of language structure.
    - **Psycholinguistics** incorporates both linguistic (competence) and psychological (performance) approaches to language.
    - **1.3.1 The Structure of Language** – To be considered a language, a communication system must:
        - Use **symbols** (sounds, gestures, written characters) representing objects, actions, events, and ideas, allowing reference to things not present.
        - Be **meaningful** and understandable by other users.
        - Be **generative**, allowing infinite messages to be produced.
        - Have **rules** governing symbol arrangement, enabling understanding of new messages.
    - **1.3.2 The Building Blocks of Language (Hierarchical)**:
        - **Phonemes**: Smallest distinguishable units in a language (e.g., 't', 'p', 'm' in English; 'o' in "bone" versus "woman").
        - **Morphemes**: Smallest meaningful units (e.g., 'I', 'a', or parts like "dis-," "-lik-," "-ed" in "disliked").
        - **Syntax**: System of rules governing how words are meaningfully arranged to form phrases and sentences (e.g., "Read the book" vs. "Read book the").
- **1.4 Language Acquisition**
    - A central topic in cognitive science, often debated between **nature vs. nurture**.
    - Current thinking: acquisition involves a **natural endowment modified by environment** (termed **innately guided learning**).
    - **1.4.1 Stages of Language Acquisition (Universal Sequence)**:
        - **Cooing**: Infant's oral exploration of vowel sounds (mostly vowel sounds), indistinguishable across babies/languages (even deaf infants). Infants can discriminate all phonetic contrasts, better than adults.
        - **Babbling**: Consonant and vowel sounds, becoming preferential production of phonemes characteristic of the infant’s own language.
        - **One-word utterances (Holophrases)**: Around 1 year, limited vowels/consonants, convey intentions, desires, demands. Vocabulary of 3 to 100 words by 18 months. Common **overextension error** (e.g., "doggie" for any four-legged animal).
        - **Two-word utterances and Telegraphic speech**: Between 1.5-2.5 years, combining single words, beginning understanding of syntax. Functional morphemes (articles, prepositions) often omitted (e.g., "want juice," "mommy sit").
        - **Basic adult sentence structure**: By about age 4, foundations of adult syntax and language structure are acquired. Vocabulary expands rapidly (300 words at 2 to 1000 at 3). By 5, most understand/produce complex sentences. By 10, language is fundamentally adult-like.

    - **1.4.2 Language Acquisition and Cognitive Science**: Scientific study began late 1950s, catalyzed by Chomsky's critique of Skinner. Chomsky argued that children learn subtle, abstract language principles without explicit instruction, suggesting an **innate, species-specific module** distinct from general intelligence.
    - **1.4.3 Language and Thought**: Debate whether language is merely labels for thoughts or shapes thinking. Whorf's hypothesis (language determines thought) is largely rejected by modern cognitive scientists. Babies can think before they talk. Cognitive psychology shows thinking in words, images, and abstract logical propositions. Children need significant **nonlinguistic cognitive machinery** to learn language.
- **1.5 Theories of Language Acquisition**

    - **1.5.1 Behaviouristic Theory (B.F. Skinner)**: Children **imitate** adult language; correct utterances are **reinforced** (e.g., praised, given what they want). No essential difference from how animals learn.
        - **1.5.2 Limitations of Behaviourism Theory**:
            - Language is rule-based; children produce novel errors (e.g., "drinked" from over-applying rules), not just imitation.
            - Children universally go through the same stages, largely unaffected by environment.
            - Children often cannot repeat adult sentences with structures they haven't acquired.
            - Parents primarily correct for truthfulness/politeness, not grammar.
            - Evidence for a **critical period**: Children not acquiring language by ~7 may never catch up (e.g., Genie).
    - **1.5.3 Innateness Theory (Noam Chomsky)**: Children are **born with an innate capacity** for language, destined to speak. They discover grammar based on inborn grammar. Universal features of language (nouns/verbs, consonants/vowels) are preordained by cognitive structure.
        - Explains rapid, perfect mastery by all children regardless of general intelligence (even severely retarded children acquire native language).
        - Process is biologically determined; brain's neural circuits contain linguistic information at birth, triggered by hearing speech. This natural faculty is the **Language Acquisition Device (LAD)**.
        - LAD contains common principles (e.g., verb tense concept); child establishes how specific language expresses these. Leads to "virtuous errors" like "drinked". Process is unconscious.
        - Slobin proposed LAD is a **process mechanism** (procedures/inference rules) for working out language rules, leading to linguistic universals as a result of innate cognitive competence.
        - **1.5.4 Evidence to Support Innateness Theory**:
            - Human anatomy is uniquely adapted for speech production (vocal tract).

        - Specific brain areas identified with linguistic functions (e.g., **Broca's area, Wernicke's area**); brain damage leads to language dysfunction.
        - Chimpanzee experiments show limited grammatical competence.
        - Formation of **creole languages**: Children of different language groups turned restricted pidgins into full, grammatically consistent languages (e.g., Dutch-based creoles in Surinam).
        - Sign languages are complex, fully grammatical languages; deaf children acquire them in similar stages to spoken language.
    - **1.5.5 Limitations of Chomsky's Theory**:
        - Work was theoretical, focused on grammar, didn't study real children or interaction.
        - Doesn't account for why a child wants to speak or the social functions of language.
        - Exposure alone is not enough (e.g., Jim, hearing son of deaf parents, needed speech therapist despite exposure to TV/radio).
- **1.5.6 Cognitive Theory (Jean Piaget)**: Language acquisition is placed within the child's **overall cognitive development**. A child must **understand a concept before acquiring the language** to express it.
    - **Piaget's Four Universal Stages of Cognitive Development**:
        - **Sensorimotor (Birth to ~2 years)**: Little competence in representing environment. Lack of **object permanence** (awareness that objects exist when out of sight).
        - **Preoperational (2 to 7 years)**: Development of language, internal representation of world (symbols, pretend play). **Egocentric thoughts** (world from own perspective). Lacks **conservation** (quantity unrelated to arrangement/appearance).
        - **Concrete Operational (7 to 12 years)**: Mastery of conservation, logical thinking, overcome egocentrism. Develops **reversibility** (e.g., clay can be rolled back). Limits to abstract thinking.
        - **Formal Operational (12+ years)**: Abstract, formal, logical thinking; hypothetical reasoning. Not all individuals reach this stage.
    - **Link to Language**: Simple ideas expressed earlier. Consistent order of mastery for function morphemes (e.g., "-ing" before plural "-s" before "to be" forms), conditioned by **logical complexity**. Object permanence relates to vocabulary increase around 18 months. Helps explain the *order* of language acquisition.
    - **1.5.7 Limitations of Cognitive Theories**:
        - Doesn't explain *why* language emerges (e.g., apes develop cognitively but not language).
        - Hard to find clear links between language and intellect beyond 18 months.
        - Syntax does not appear to rely on general intellectual growth.

    - **1.5.8 Input or Integrationist Theories (Jerome Bruner)**: Emphasize importance of **language input from caregivers**. Language learned in context of interaction.
        - **Child-directed speech (CDS)**: Adults adapt their speech to children to support acquisition (termed **scaffolding**).
        - Bruner coined **Language Acquisition Support System (LASS)**, as a complement to Chomsky's LAD.
        - Turn-taking in conversation develops non-verbally before words.
        - Children hear sentences in context, inferring meaning from context and word knowledge.
        - **1.5.9 Limitations of Input Theories**: Useful corrective, but universal stages suggest CDS may be useful but not essential, as some cultures don't adopt special ways of talking to children.
        - No single theory fully explains the process; each offers a partial explanation.
- **1.6 The Biology of Language Acquisition**
    - Human language enabled by special adaptations of the human mind and vocal anatomy, evolved for speech.
    - Human vocal tract allows precise articulation (low larynx, right-angle bend).
    - **Chimpanzees vs. Humans**: Chimps acquire rudimentary hand-signs/commands with massive teaching, mostly for requests. Humans spontaneously acquire thousands of words, combine them structurally, and use sentences for varied purposes. This "lack of homology" (difference) doesn't refute Darwinian evolution; both derived from a common ancestor 6-7 million years ago, allowing 300,000 generations for human language evolution. Language likely evolved for sharing knowledge and cooperation.
    - **1.6.1 Maturational Changes in Brain**: A driving force for language acquisition.
        - Brain development post-birth: rapid increase in head size, brain weight, cerebral cortex thickness, and synapses (peaking 9 months-2 years). Long-distance connections myelinate throughout childhood. Metabolic activity peaks around age 4.
        - These changes may be minimum levels for babbling, first words, and grammar.
        - May explain **decline in language learning ability** over lifespan.
        - Brain language circuitry is more **plastic in childhood** (children learn/recover language after left hemisphere damage better than adults).
        - **Critical period** for successful language acquisition: typically by 4, guaranteed up to 6, steadily compromised until puberty, rare after. Linked to neurological changes (e.g., metabolic rate decline, synapse changes).
    - **1.6.2 Dissociations Between Language and General Intelligence**:
        - Humans evolved dedicated brain circuitry for language, mostly in the left hemisphere (sylvian fissure).

        - Strokes can cause catastrophic language loss (aphasia) without impairing other intelligence aspects (e.g., nonverbal IQ).
        - Syndromes show the opposite: intact language with severe retardation (e.g., **Williams Syndrome**: average IQ ~50, daily task incompetence, but fluent, articulate language). This indicates language development is **not dependent on fully functioning general intelligence**.
    - **1.6.3 Neural Networks**: Computer models that learn language statistical structure from examples, not preprogrammed with rules. Developers speculate children may acquire language similarly through exposure.

Unit 2: Language Processing (Comprehension and Language Expression)

- **2.0 Introduction**
    - Language is the **principal means to acquire and express knowledge**, making its study central to cognitive psychology.
    - Language processing is a dynamic process, involving microstructure analysis in real time.
    - Approaches to understanding language: acoustic peculiarities, grammatical structure, psycholinguistic processes at discourse level.
- **2.2 Functions of Language**
    - Primarily for **communication** and conveys **meaning**.
    - Conveys **intentions, motives, feelings, and beliefs**.
    - Used to **issue requests and commands, teach, and convey information**.
    - Can represent **abstract ideas** (e.g., beauty, justice) and events not tied to the present (symbolic).
- **Elements of Language Expression (Speaker-Listener Situation)**:
    - **Speech Acts**: Speakers intend to influence listeners by conveying intentions (e.g., assertions, verbal commitments, warnings, commands). Can be **direct** or **indirect**, with meaning depending on context.
    - **Propositional Content**: Ideas conveyed in a sentence. A sentence can express multiple propositions (e.g., "The bright student received an A in Mathematics" contains "student is bright" and "student received A"). More propositions lead to longer reading time.
    - **Thematic Structure**: Speaker pays attention to listener's knowledge and understanding to communicate effectively.
- **2.3 Structure of Language**
    - Language divided into three basic parts: **phonology, syntax (grammar), and semantics**.
    - **2.3.1 Basic Units of Language**:

        - **Phonemes**: Rules for pronunciation of speech sounds. English has between 15 and 85 phonemes.
        - **Morphemes**: How these sounds combine to produce meaningful units.
    - **2.3.2 Higher Levels of Linguistic Analysis**:
        - **Phonology**: Study of speech sounds.
        - **Morphology**: Study of how sounds combine to produce morphemes.
        - **Lexical Content**: Analysis of what words are used and their frequency; predicts ease of word learning.
        - **Syntactic Content**: Focus on arrangement/ordering of words to form phrases and sentences (grammar rules).
        - **Semantic Content**: Focus on the meaning of a passage or sentence. Listener actively formulates hypotheses based on context.
    - **2.3.3 Phrase Structure of Sentences**: Sentences are organized into phrases (groupings of words), typically a **noun phrase** and a **verb phrase**, which have subcomponents. Pauses in speech usually reflect underlying phrase structure.
    - **2.3.4 Surface Structure and Deep Structure**:
        - **Surface Structure**: The organization of phrases as a sentence is actually spoken or read, reflecting its phonological realization.
        - **Deep Structure**: The underlying structure that includes the abstract meaning of a sentence.
        - Sentences with the same meaning but different surface structures (e.g., "Rohit threw the ball" and "The ball was thrown by Rohit") have the **same deep structure**.
        - Sentences with the same surface structure but **different deep structures are ambiguous** (e.g., "The lamb is ready to eat," can mean the lamb is food or is prepared to eat).
        - **Transformational rules** (Chomsky) link deep structure to surface structure. Memory stores an underlying base structure (schema) plus rules to regenerate complex sentences.
        - Language is comprehended in syntactically and semantically defined chunks.
- **2.4 Processes in Language**

    - Includes production, speech perception/comprehension, and language development.
    - **2.4.1 Production of Language (Speaking)**:
        - Begins with **planning** based on intended effect on listener, speaker's knowledge of listener's understanding, and desired syntactic/semantic/pragmatic form.
        - It is an **instrumental act** (to produce an effect).
        - **Five Steps (Clark & Clark, 1977)**:
            - **Discourse Plans**: Deciding the type of discourse (conversation, description, joke, instructions), which has a particular structure.

        - **Planning Sentences**: Selecting specific sentences to meet objectives, determining speech act, propositional content, and thematic structure (e.g., describing a house from global to specific details).
        - **Constituent Plans**: Planning specific components of the sentence (words, phrases, ordering).
        - **Articulatory Program**: Planning the physical execution of speech (coordinated muscular contractions).
        - **Articulation**: The actual output of speech.
    - **Gricean Maxims** for good communication: **Quantity** (avoid rambling), **Quality** (don't lie), **Relation** (be relevant). Violations are informative.
    - **Slips of the tongue** reveal regularities in planning (e.g., morphemes slip as entire units, seldom "illegal" combinations).
- **2.4.2 Speech Perception and Comprehension**:
    - Humans perceive speech rapidly (e.g., 50 phonemes per second for fluent speakers).
    - Involves complex processes of encoding and comprehension, not just simple sound identification. Interpretation, meaning, and contextual influences play important roles.
    - **Context Effect**: Word recognition is highly dependent on context and knowledge. Context allows inference (e.g., "frightening d...." allows inference of "dream").
    - **Phonemic Restoration**: Subjects "restore" missing speech sounds based on context, unable to locate where the sound was obscured (e.g., cough masking an 's' sound). Indicates an active word-searching process.
    - Spoken words are not perceived as distinct units; often incomprehensible out of sentence context. Speech is comprehended based on many factors beyond sounds themselves (intentions, context, expectations).
    - **Theories**:
        - Some theories equate speech perception with auditory perception (template-matching or feature-detection) (e.g., Phonetic refinement theory, TRACE model). TRACE model proposes interactive levels (acoustic, phoneme, word).
        - Cognitive and contextual factors influence perception; perceived speech may differ from actual sounds.
- **2.5 Language Development (Semantics)**:
    - Follows an orderly course (as described in Unit 1).
    - Sounds acquire meaning by becoming associated with objects, symbols, events (e.g., "Mama" with mother).
    - Children learn **semantic features** and may **overgeneralize** (e.g., "ball" for moon, orange), then construct complex sentences.
    - Children acquire **linguistic competence** (knowledge to produce grammatically correct sentences) even if they can't verbalize rules.

        - **Overgeneralization** (e.g., "goed" after learning "-ed" rule, then returning to "went") is strong evidence of learning syntactic rules.
        - Understanding of **speech acts** (e.g., "I promise") develops with age and depends on grasping contextual circumstances.
- **2.6 Comprehensive Model of Language Processing (Kintsch's Model of Comprehension)**

    - An influential and comprehensive model.
    - Integrates memory and comprehension of written/spoken language.
    - Combines **top-down processing** (goal schema determines relevant material) and **bottom-up processing** (stimulus components lead to whole recognition).
    - Based on **propositions**: Abstractions from observations (reading/listening) that are retained in memory. Each proposition consists of a **predicate** (verbs, adjectives, adverbs) and one or more **arguments** (nouns, noun phrases, clauses).
    - **Processing flow**: Initial text processing occurs in **Short Term Memory (STM)** (limited capacity). If new and old propositions don't match in STM, a **reinstatement search** in **Long Term Memory (LTM)** occurs, which can make text difficult. Readers make inferences to connect ideas.
    - **Propositional Representation of Text and Reading**: Sentences with more propositions are harder to comprehend, even if surface complexity is similar. Research shows a consistent relationship between the number of propositions and reading time.

Unit 3: Multilingualism and Cognition

- **3.0 Introduction**

    - **Multilingualism** is a natural human potential; monolingualism is often due to environmental factors.
    - Chomsky's **Language Acquisition Device** and **Universal Grammar** (Principles and Parameters) provide an innate basis for language acquisition, influencing how multiple languages are acquired.
- **3.2 Multilingualism – Basic Concepts**

    - **Definition**: Knowledge of more than one language by a person or social group, with the ability to switch between them.
    - **Terms**: Bilingualism, polylingualism, plurilingualism, diglossia, languages-in-contact. A multilingual person is a **polyglot**.
    - **Types**:
        - **Personal Multilingualism**: Individual knowledge/behavior, not necessarily community-wide.
        - **Social Multilingualism**: Communicative practices of a group sustaining multiple languages (e.g., India with ~200 languages).

    - **3.2.1 The Structure of Multilingualism**: A broader definition acknowledges regular use and switching, regardless of symmetry of command, acquisition modality, or language distance.
    - **Kinds of Bilingualism**:
        - **Elite bilingualism**: Language learned formally (e.g., school).
        - **Neighborhood bilingualism**: Language acquired naturally through interaction.
        - **Additive bilingualism**: Second language acquired *in addition* to a well-developed first language.
        - **Subtractive bilingualism**: Elements of a second language *replace* elements of the first language.
        - **Simultaneous bilingualism**: Child learns two languages from birth.
        - **Sequential bilingualism**: Individual learns one language, then another.
    - Babbling onset is similar for children exposed to one or two languages. Multilingualism is common globally (e.g., India, Belgium), affecting 60% of the world's population. Language and culture are interconnected.
- **3.3 Multilingualism and Cognition**
    - Multilingualism provides rich information about language organization and processes.
    - **Key Issues**:
        - **Code-switching**: Multilinguals' ability to select words from either language within a sentence. Maximizes communication between bilinguals. Often requires no additional time and may be faster than single-language usage.
        - **Lexical vs. Semantic Knowledge**:
            - **Concept Mediation account**: Lexical knowledge in language-specific memory, but **semantic knowledge in a common conceptual memory store**. Supported by evidence, especially in fluent multilinguals.
            - Alternative view: both types of knowledge in language-specific stores.
        - **Language Familiarity and Voice Identification**: Bilinguals show consistent correct voice identification across languages/accents, an advantage over monolinguals who are best at their native language.
    - **3.3.2 Rule-governed Language Choice**: Multilinguals' choice of language is not arbitrary but governed by rules, often based on communication domains (private/public, professional/leisure).
    - **3.3.3 Mixing is Rule-governed Too**: Multilinguals can choose a **monolingual mode** (other language 'switched off') or a **bilingual mode** (entire repertoire activated). Bilingual mode requires mutual agreement. Rules and norms govern 'grammatical' mixing of elements from different languages. The **matrix language** provides cognitive scaffolding, and speakers switch to an **embedded language**

for missing words or specific effects, provided the word "fits" the grammatical slot (otherwise, an "embedded language island" may occur).

- **3.4 Multilingualism and Thinking**
    - Debate on whether multilinguals think differently from monolinguals; data is contradictory, influenced by participant background (e.g., balanced vs. unbalanced bilinguals, middle-class).
    - **Additive Bilingualism** (L2 in addition to L1) tends to **increase thinking ability**.
    - **Subtractive Bilingualism** (L2 replaces L1) tends to **decrease thinking ability**.
    - There may be a **threshold effect**: a certain high level of competence in both languages is needed for positive effects.
    - **3.4.1 Other Benefits of Multilingualism**:
        - **Enhanced metalingual abilities**: Keener awareness and sharper perception of language (e.g., detecting anomalous sentences).
        - Improved understanding of **how language works** and ability to manipulate language for thinking/problem solving.
        - Better ability to **deal with distractions**, potentially offsetting age-related mental decline.
        - Better **listening skills and sharper memories**.
        - Development of **verbal and spatial abilities**.
        - Greater **cognitive flexibility**, better problem solving, and higher-order thinking skills.
        - Improved conceptual development, creativity, analogical reasoning, divergent thinking, and figural creativity.
        - More accustomed to switching thought patterns, leading to **more flexible minds**.
        - Outperform monolingual peers in core subject areas on standardized tests.
        - Possess extra skills in language use: transfer, borrowing, various forms of code-switching, mixing, and translation patterns.
        - Overall, people who know more than one language usually **think more flexibly** than monolinguals.
- **3.5 Acquisition of a Second Language**
    - Acquired at different ages, situations, and competence levels. Social context is crucial. Passive exposure in childhood can lead to reactivatable competence.
    - **Age**: Vocabulary comprehension and fluency can be acquired well after adolescence. However, **nativelike pronunciation depends on early acquisition**. No critical periods for L2 acquisition, with the exception of accent.
    - Adults may find L2 harder if they retain L1 as dominant; young children may switch dominance.
    - Process involves cognitive procedures, building new grammar, assisted by a "language acquisition device" (less for L2) and a **Language Acquisition Support System (LASS)** based on social interaction.

    - Effective learning experiences depend on individual cognitive abilities and the specific context/uses for the L2.
    - L2 learning difficulty varies by language (e.g., Spanish easier than Russian for English speakers due to shared roots and inflection). L1 difficulty does not predict L2 difficulty.
    - **3.5.1 Single-System Versus Dual-System Hypotheses**: How are different languages represented in the mind?
        - **Single-System Hypothesis**: Two or more languages are represented in **one system or brain region** (supported for early multilinguals).
        - **Dual-System Hypothesis**: Different languages are represented in **separate systems of the mind**.
        - **Evidence from Brain Damage**: Studies are equivocal; sometimes L1 recovers first, sometimes L2, sometimes mixed, sometimes independently. This suggests **at least some duality of structure**.
        - **Electrical Stimulation Studies**: Mild electrical stimulation inhibited object naming in both languages for bilingual patients, suggesting **some aspects of two languages may be represented singly, while others separately**.
- **3.6 Neural Mechanism of Multilinguals**
    - Neuroscience (e.g., fMRI) investigates differences in brain networks.
    - **Early bilinguals** seem to build a sufficiently adaptable network to integrate later acquired languages.
    - **Late bilinguals** may establish new neural areas for their later-acquired languages.
    - Suggests multilingual repertoires and their use are the **default case**, with monolingual competencies as just one variation.

Unit 4: Language and Speech Disorders

- **4.0 Introduction**
    - Normal language acquisition is sequential (vocalization, speech sounds, prosodies) and involves recognized stages for **form** (phonology, morphology, syntax), **content** (semantics), and **use** (pragmatics).
    - Disorders occur when this natural order is disrupted.
- **4.2 Defining Language and Speech Disorders**
    - Disordered if they differ from the cultural norm. Cultural practices (e.g., verbal hesitancy in some Asian cultures) should not be confused with disorders.
    - Often **not physically visible** until the person speaks.
    - Causes: physical, mental, or socialization defects.
- **4.3 Language Disorders**

- **Language** is the rule-based use of speech sounds to communicate.
- **Language disorders** or impairments involve the **processing of linguistic information** (grammar, semantics, other aspects).
- Can be **receptive** (difficulty understanding speech) or **expressive** (difficulty producing speech sounds according to rules) or both. Affect spoken, written, and sign language; typically, all forms are impaired.
- **Characteristics**:
  - Use of speech sounds in patterns that fail to follow arbitrary language rules (e.g., lack of communication etiquette like talking out of turn).
  - Delay in speech sound use relative to normal physical, cognitive, and social development.
  - Breakdown in development or use of language knowledge on the usual schedule.
- **Specific Language Disorders**:
  - **4.3.1 Aphasia**: Impairment of language functioning caused by **damage to the left hemisphere of the brain**.
    - **Wernicke's Aphasia**: Damage to the **left temporal lobe**. Characterized by **notable impairment in understanding spoken words/sentences**. Speech is generally **fluent but semantically incoherent** (uses nonsense words, improper substitutions, sentences that make no sense, e.g., "the scroolish prastimer ate my spanstakes"). Treatment often involves supporting nonlanguage communication.
    - **Broca's Aphasia**: Damage to the brain's **premotor area** (controls speech production). Characterized by **nonfluent, effortful speech**. Patients often omit function words (e.g., "a", "the") and produce ungrammatical sentences. They have **difficulty using syntactic information** for understanding sentences (e.g., struggle with "the dog that the woman is biting is grey" because it requires syntactic analysis, not just real-world knowledge). Differs from Wernicke's: speech is agrammatical, but verbal comprehension is largely preserved.
    - **Global Aphasia**: Combination of **highly impaired comprehension and production of speech**, caused by lesions to both Broca's and Wernicke's areas.
    - **Anomic Aphasia**: Difficulties in **naming objects** and retrieving words. May be specific to certain categories (e.g., living things).
    - Aphasias suggest that phonological, syntactic, and semantic language information are stored and processed separately in the brain.
  - **4.3.2 Autism**: Developmental disorder with abnormalities in social behavior, language, and cognition. Biological origins.
    - **Symptoms**: Identified around 14 months (failure to show normal social interaction), repetitive movements, stereotyped interests.

About half fail to develop functional speech. Common is **echolalia** (repeating speech heard, sometimes hours later).
        - Associated with brain abnormalities. More common in males (5x). Increased diagnosis may be due to diagnostic changes or environmental pollution.
- **4.3.3 Learning Disability (Language-based)**: Problems with age-appropriate reading, spelling, and/or writing. Most affected individuals have average to superior intelligence.
    - **Dyslexia**: Specific learning problem of reading. Not due to mental retardation, brain damage, or lack of intelligence. Caused by impairment in brain's ability to translate visual/auditory images into understandable language. Severity varies; earlier treatment is better, but never too late.
    - **Warning Signs**: Letter/number reversals, difficulty copying, uncoordinated, difficulty with organized sports, difficulty with left/right, auditory problems (remembering/understanding what is heard, missing parts of words/sentences, trouble finding words).
    - **Types of Dyslexia**:
        - **Primary Dyslexia**: Dysfunction of left cerebral cortex, doesn't change with age, hereditary.
        - **Secondary/Developmental Dyslexia**: Caused by hormonal development in fetus, diminishes with maturity.
        - **Visual Dyslexia**: Number/letter reversals, inability to write symbols in correct sequence.
        - **Auditory Dyslexia**: Difficulty with sounds of letters/groups, perceived as jumbled.
        - **Dysgraphia**: Difficulty holding/controlling a pencil for correct markings.
    - **Impact**: Frustration, withdrawal, depression, low self-esteem, strained social interactions, unmotivated, dislike for school.
- **4.3.4 Alzheimer's Disease**: Brain disorder causing **decrement in language processing ability**. Primarily afflicts elderly, causing progressive, irreversible damage to cortical regions, impacting memory.
    - **Language characteristics**: Striking simplification; precise words are lost and replaced by **more general terms** due to fundamental loss of categorical organization in semantic memory. Phonological and syntactic knowledge/use are minimally affected. Deficits in **pragmatic knowledge** (e.g., recognizing speech act intentions, turn-taking). This pattern supports the distinctiveness of language information levels.
- **4.3.5 Specific Language Impairment (SLI)**: Developmental language disorder in absence of other neurological, sensorimotor, nonverbal cognitive, or social-emotional deficits. Children lag peers in language production/comprehension, contributing to learning/reading disabilities.

        - **Hallmarks**: Delay/deficit in use of **function morphemes** (e.g., "the," "a," "is") and other **grammatical morphology** (e.g., plural "-s," past tense "-ed"). Problems with phonological awareness (combining/selecting speech sounds). Also associated with impoverished vocabulary, word finding problems, difficulty learning new words.
        - Causes: Delays in acquiring specific linguistic mechanisms (e.g., verb tense rule), or deficits in processing brief/rapidly changing auditory information, or remembering temporal order of auditory info. Poor short-term memory for speech sounds correlates with vocabulary acquisition. About 50% of children with SLI develop reading difficulties/dyslexia.
    - **4.3.6 Developmental Phonological Disorders**: Affect children's ability to develop easily understood speech by age four; can affect reading and spelling. Difficulty learning and organizing all sounds needed for clear speech. May co-occur with immature grammar, stuttering, or word-retrieval issues. Cause largely unknown, but may have a genetic component. Speech-language pathology treatment is usually successful.
    - **4.3.7 Dyspraxia (Developmental Dyspraxia)**: Impairment in the ability to **plan and carry out sensory and motor tasks**. Individuals may appear "out of sync" with their environment.
        - **Symptoms**: Poor balance/coordination, clumsiness, vision problems, perception difficulties, emotional/behavioral problems, difficulty with reading, writing, speaking, poor social skills, poor posture, poor short-term memory. May have average/above average intelligence but behave immaturely.
        - Lifelong disorder, but can be compensated through occupational and speech therapy.
- **4.4 Speech Disorders**
    - Characterized by difficulty in **producing normal speech patterns**. What's normal for one age may indicate a problem in an older child.
    - Speech is disordered in three ways: **voice, articulation, and fluency**.
    - **Types of Speech Disorders**:
        - **4.4.1 Voice Disorders**: Abnormalities in pitch, volume, vocal quality, resonance, or duration of sounds.
            - **Incorrect phonation**: breathy, strained, husky, hoarse voice.
            - **Incorrect resonation**: hyper-nasality or hypo-nasality.
            - Can be due to improper voicing habits.
            - Paralanguage features (pitch, volume, intonation) are culturally determined.
        - **4.4.2 Speech Sound Disorder (Articulation Disorders)**: Difficulty producing specific speech sounds (often consonants like /s/ or /r/).

        - **Articulation Disorders (Phonetic)**: Difficulty *physically producing* sounds. Sounds are added, omitted, substituted, or distorted. Causes: structural abnormalities (e.g., cleft lip/palate, tongue-tie, missing teeth), faulty learning, nervous system damage. Can affect self-concept.
        - **Phonemic Disorders**: Difficulty learning **sound distinctions** of a language (one sound used in place of many). Individuals have trouble distinguishing sounds, leading to consistent incorrect substitutions. Usually improve with speech therapy.
    - **4.4.3 Fluency Disorders**: Impairment in the normal rate or rhythm of speech.
        - **Cluttering**: Very rapid speech with extra sounds.
        - **Stuttering**: Sounds repeated or blocked, especially at word beginnings. More prevalent in children, due to familial, psychological, neurological, and motoric factors. Adversely affects socialization.
    - **4.4.4 Apraxia of Speech (Verbal Apraxia/Dyspraxia)**: Trouble saying what one wants correctly and consistently. Severity varies.
        - **Acquired Apraxia of Speech**: Occurs at any age (typically adults), caused by **brain damage** (stroke, head injury, tumor), involves loss of *existing* speech abilities. May co-occur with dysarthria or aphasia.
        - **Developmental Apraxia of Speech (DAS)**: Present from birth in children (more boys than girls). Different from developmental speech delay. Causes unknown, possibly related to overall language development or neurological signals to speech muscles. Genetic factors suggested.
        - **Symptoms**: Difficulty ordering sounds/syllables, inconsistent mistakes, groping for words, incorrect use of **prosody** (varying rhythms, stresses, inflections).
    - **4.4.5 Dysprosody**: Rarest neurological speech disorder. Alterations in intensity, timing, rhythm, cadence, and intonation of words. Caused by brain lesions (vascular accidents, trauma, tumors).
    - **4.4.6 Dysarthria**: Motor speech disorder. **Weakness or paralysis of speech muscles** caused by damage to nerves and/or brain. Causes: strokes, Parkinson's, ALS, injuries, cerebral palsy.
        - **Symptoms**: Slurred speech, speaking softly or mumbling, slow/rapid speech, limited tongue/lip/jaw movement, abnormal intonation, nasal/stuffy vocal quality, hoarseness, breathiness, drooling, chewing/swallowing difficulty.
- **Treatment**: Speech-language pathologist (SLP) evaluates.
    - **Articulation therapy**: For isolated speech disorders (practicing sounds/words).

        - **Fluency training**: For stuttering/fluency disorders (coordinating speech/breathing, slowing rate, prolonging syllables).
        - **Delayed Auditory Feedback (DAF)**: Effective for stuttering.
        - **Neurodevelopmental approach**: For severe/multiple disabilities (inhibits reflexes for normal movement).
        - **Other techniques**: Motor-kinesthetic approach, biofeedback.
        - **Alternate communication**: For severe disorders (manual signing, computer-synthesized speech).

Block 4: Problem Solving

Unit 1: Nature of Problem Solving

- **1.0 Introduction**
    - **Problem Solving** involves finding a way out of difficulty, around an obstacle, or attaining a goal not immediately understandable. It's a fundamental human activity, nearly inescapable in daily life.
    - Problems vary from finding a meal to developing a psychological test.
- **1.2 Types of Problems**
    - **Well-defined problems**: Both the nature of the problem and information needed to solve it are clear (e.g., mathematical equation, jigsaw puzzle).
    - **Ill-defined problems**: Specific nature of the problem and information needed are unclear (e.g., how to bring peace).
    - **Greeno's (1978) classification for well-defined problems**:
        - **Arrangement problems**: Require rearranging or recombining elements to satisfy criteria (e.g., anagrams like NEWDOOR to ONEWORD, jigsaw puzzles).
        - **Inducing Structure problems**: Must identify existing relationships among elements, then construct new relationships (e.g., analogies, series completion problems).
        - **Transformation problems**: Change an initial state to a goal state using specific rules (e.g., Tower of Hanoi). Primarily requires **planning skills based on means-end analysis** (reducing differences between current and goal state).
- **1.3 Characteristics of Difficult Problems**
    - **Intransparency**: Lack of clarity.
    - **Commencement opacity**: Confusion on how to start.
    - **Continuation opacity**: Ongoing confusion during the process.
    - **Polytely**: Multiple goals, making goal selection difficult.

    - **Inexpressiveness**: Inability to clearly state the problem.
    - **Opposition**: Conflicting elements.
    - **Transience**: The problem keeps changing.
    - **Complexity**: Too many items, interrelationships, and decisions.
    - **Enumerability**: Cannot be easily listed or quantified.
    - **Connectivity**: Hierarchy of related problems.
    - **Heterogeneity**: Problem is not homogeneous.
    - **Dynamics**: Time considerations.
    - **Temporal constraints**: Time limitations.
    - **Temporal sensitivity**: Problem is affected by time.
    - **Phase effects**: Changes in different phases of the problem.
    - **Dynamic unpredictability**: High degree of unpredictability.
    - Resolving difficult problems requires a direct attack on these characteristics.
- **1.4 Nature of Problem Solving**
    - **Study Methods**: Introspection, behaviorism, simulation, computer modeling, experiment.
    - Early research used simple, novel lab tasks (e.g., Tower of Hanoi), assuming they reflected "real-world" problems. Such tasks simplify logic/reasoning steps, but omit complexity/emotional aspects of real-world problems.
    - Human problem solving involves two processes: **problem orientation** (motivational/attitudinal approach) and **problem-solving skills** (cognitive behavioral steps).
    - It is a **mental process**, part of problem finding and shaping. Considered a **higher-order cognitive process**.
    - Occurs when moving from a given state to a desired goal state. It is a form of **thinking or cognition**.
    - **Cognition**: Mental representation of information that can be manipulated to solve problems.
    - Problem solving is necessary when a goal is blocked by obstacles.
    - **1.4.1 The Stages of Problem Solving (Molar Components)**:
        - **The initial state**: How the starting conditions are defined.
        - **The Operators**: Moves or operations to transition from one state to another.
        - **Intermediate Problem States**: Any states generated by applying an operator on the way to the goal.
        - **The goal state**: How the final state or goal conditions are described.
        - **Problem Space**: The internal (mental) representation of these four states of a problem; unique to each individual and dependent on the problem's nature. The initial state is critical.
- **1.5 Types of Thinking Involved in Problem Solving (Wertheimer, 1959)**
    - Effective problem solving requires:

        - **Productive thinking**: Grasping general principles, restructuring the problem, going beyond previously learned solutions.
        - Being sensitive and open to structural requirements.
    - **1.5.1 The Kinds of Thinking Processes**:
        - **Analytical Thinking**: Solution is contained within the premise (e.g., simple counting like "how many doors in your house").
        - **Synthetic Thinking**: Conclusion is not in the premise; involves constructing a **mental model** (e.g., images, number lines) to derive new solutions by seeing new relationships. This is also termed **insight**.
        - **Newell's Mental Model**: Goal is to construct a mental model, then find answers by inspecting it ("mind's eye"). The "mind's eye" concept is controversial in cognitive psychology.
- **1.6 Concept of Insight Problems**

    - **Insightful solutions**: Appear suddenly, without warning, characterized by "sudden illumination".
    - **Non-insight solutions**: Solved gradually, through a stepwise progression (e.g., arithmetic, algebra).
    - **Feeling of Warmth**: Subjects solving non-insight problems report feeling "warmer" (closer) to the solution as they progress. For insight problems, this feeling only appears when the solution suddenly arises.
    - **Metacognition**: Judgments about one's own knowledge. Metacognitive assessments are accurate for non-insight problems but not for insight problems.
    - **Source of Difficulty**: Often the inability to recognize that existing knowledge is needed. Hints can be effective. People struggle to apply old information to new problems.
    - **Sagacity**: The ability to select relevant information and discern what is important in a situation.
    - **Information Format**: Presenting information in a **puzzle format** helps discriminate relevant information better than a declarative format, making it more accessible for later use.
    - **Example**: The "lake" riddle (e.g., "stone rested on the surface of the lake for 3 months, then sank") requires accessing the less common meaning of "lake" (frozen) for insight.

Unit 2: Stages of Problem Solving

- **2.0 Introduction**: This unit examines the steps and strategies to reach solutions.

- **2.2 Nature of the Problem**: The first step is to **understand the problem's aspects, obstacles, and goals**. This involves using background knowledge to distinguish important from irrelevant information and representing the problem effectively (e.g., using symbols, lists, matrices, graphs, visual images).

- **2.3 Generating Solutions**: The next stage is to **find a variety of potential solutions** using past experiences and present resources.

- **2.3.1 Judging the Best Among the Likely Solutions**: Evaluate each alternative for outcomes, obstacles, and costs, then choose the best solution.

- **Carrying out the best solution**: Implement the chosen solution. Strategies include algorithms and heuristics.

- **Evaluation of solution**: Compare the solution with the problem representation and check if the path taken was legal. If inadequate, restart the process.

- **2.3.2 Strategies for Problem Solving**:

    - **Algorithm**: A **specific procedure guaranteeing a correct answer** if followed (e.g., rules of multiplication).
        - **Systematic random search**: Tries all possible answers in an orderly way (e.g., unscrambling YBO to YOB, BYO, BOY). Can be time-consuming for many alternatives.
        - **Unsystematic random search**: Tries possible answers randomly without record, may repeat attempts.
    - **Heuristics**: **Rules of thumb** that simplify problems and allow rapid solutions, but **do not guarantee a correct answer**.
        - Example: For an anagram like DWARG, look for familiar letter combinations (dr, gr) to quickly find "draw," "drag," "grad". Drawback: might miss some words.
        - Psychologists focus more on human use of heuristics.
- **2.3.3 Some Specific Techniques of Problem Solving (General, Domain-Independent)**:

    - **Generate and Test**: Generate possible solutions, then test if they meet criteria (e.g., listing edible things starting with 'c' and discarding non-eatables). Effectiveness decreases with many possibilities or lack of guidance. Useful when possibilities are few (e.g., finding lost keys in a small area).
    - **Means-ends Analysis**: Divide problem into **subproblems**. Reduce the difference between the current state and the goal state (e.g., solving an equation by performing operations to isolate 's'). Involves repeatedly setting and working towards subgoals.
    - **Backward Search**: Start at the **goal state** and work backward to the initial state. Useful for problems with a uniquely specified goal state (e.g., mazes with many dead ends).
    - **Planning Strategy (Analogy)**: Divide problem into simple and complex aspects; solve the simple first, then the complex. **Analogy** involves using a solution from a

previously solved similar problem (e.g., solving BODUT by thinking of familiar D, O, U combinations).
    - **Thinking Aloud (Concurrent Verbalization)**: Verbalizing thought processes while solving a problem. This provides a **protocol** (verbal description) of the solution process. Newell (1977) steps include dividing protocol into phrases and constructing a problem behavior graph.
        - **Limitation**: May interfere with non-reportable or unconscious processes leading to insightful solutions (e.g., memory searches). Silent control groups are recommended in research.
    - **Other Strategies**: Abstraction (solve in a model), Divide and conquer (break into smaller problems), Hypothesis testing (assume explanation), Lateral thinking (indirect solutions), Method of focal objects (synthesize non-matching characteristics), Reduction (transform to a known problem), Research (adapt existing solutions), Root cause analysis (eliminate cause), Trial-and-error, Brainstorming (group ideas).
- **2.4 Creativity and Problem Solving**
    - **Creativity** is closely linked to **insight**. It involves **appropriate novelty** (originality that suits the context). Seen as a **combination or recombination of knowledge**.
    - **Insight Problem-Solving Skills (Davidson & Sternberg)**:
        - **Selective Encoding**: Determining which information is relevant amidst useless data (e.g., Fleming recognizing the bacteria-killing ability of mold for penicillin).
        - **Selective Combination**: Discovering a novel way to combine elements of a solution (e.g., Darwin combining natural history facts into the theory of evolution).
        - **Selective Comparison**: Discovering a novel comparison that leads to unforeseen consequences (e.g., Kekule discerning the benzene ring structure from a dream of a snake biting its tail).
    - Insight is a key aspect of the creative process where a solution suddenly comes to mind.
    - **2.4.1 Stages Involved in Creative Discoveries**:
        - **Preparation**: Setting a goal and gathering relevant information.
        - **Incubation**: A period of apparent inactivity, potentially involving unconscious mental activity or waiting for a missing link.
        - **Insight (Illumination)**: The moment when familiar elements combine in new, useful ways.
        - **Verification**: Testing if the new combination is, in fact, useful.
    - **Characteristics of Creative Individuals**: A clear sense of purpose and a strong store of knowledge and techniques in their field.
- **2.5 Artificial Intelligence (AI) in Problem Solving**

- **Problem Space Hypothesis**: Every possible state within a problem corresponds to a mental node, and the entire set of nodes forms a "problem space".
- Used to create **expert systems**: Computer programs designed to model human expert judgments. They contain a **knowledge base**, **inference rules**, a **search engine**, and a **user interface**.
- AI approaches involve computer programs that solve problems in ways similar to intelligent human problem solving.
- **2.5.1 Criticisms of the Computer Simulation Approach**:
  - Some argue human thinking is too complex for machines to mimic. (Counter: programs solve complex problems, no inherent reason they can't match/surpass human complexity).
  - Philosophical objection: computer programs don't "really" think; the essence of thinking lies beyond reasoning/calculating (Heidegger). (Counter: AI is widely used and promising).

Unit 3: Theoretical Approaches to Problem Solving

- **3.0 Introduction**
  - Problem solving varies in consciousness, goal definition (well-defined vs. rambling), and knowledge required (knowledge-lean vs. knowledge-rich).
  - Early research focused on knowledge-lean puzzles; later research considered knowledge-rich expert problem solving.
- **3.2 Approaches to Problem Solving**
  - **3.2.1 Traditional Approaches**: Explain problem solving using principles of **associative learning** (classical and instrumental conditioning). Prior experience creates stimulus-response associations; problem difficulty depends on the strength of correct associations. Stresses **transfer of prior learning** and learning during problem solving.
  - **3.2.2 Gestalt Approaches**: Emphasize the **importance of problem structure** and the formation of **new combinations of old ideas**. Interested in how people solve problems by **rearrangement of objects** (e.g., Kohler's ape moving sticks/crates to reach fruit). Solutions often occur as a **flash of insight** (sudden discovery after incorrect attempts).
    - Distinguished between:
      - **Reproductive thinking**: Application of tried-and-true paths, using rote memory.
      - **Productive thinking**: Requires **insight and creativity**, seeing a new way to organize or structure problem elements (e.g., the nine-dot problem, where solution requires drawing lines outside the perceived square).
  - **3.2.3 Information Processing and Computer Simulation Approach**: Programs computers to perform tasks like humans. Compares computer performance to

human subjects. Views problem solving as processing, modifying, and transforming information, and retrieving factual/procedural knowledge from long-term memory.

- **3.2.3.1 The General Problem Solver (GPS) (Newell, Shaw, Simon, 1958)**: Influential theory.
  - Equipped with a limited capacity **working memory**, large capacity **long-term memory**, **serial processor**, and reliance on **heuristics**.
  - Used **verbal protocols** (records of people thinking aloud) to identify general heuristics.
  - Assumes problem solver represents problem as a **problem space**: a set of nodes (knowledge states). The solver moves from an initial state to a goal state by applying **operators** (permitted actions). Problem solving is a **constructive search** to build this path.
- **3.2.3.2 Means-End Analysis (Heuristic)**: A general problem-solving heuristic to **reduce the difference between the current knowledge state and the goal state**.
  - Steps: Set a goal, find difference, find method to decrease difference, set method application as a subgoal, apply means-end analysis to subgoal.
  - Involves setting up goals and subgoals. Can be expressed as a **production system** (if-then rules).
- **3.2.3.3 Wickelgren's General Problem Solving Strategies**: Based on information processing theories like GPS.
  - A formal problem contains: initial state, goal state, set of operations/transformations.
  - A solution is a sequence of states or actions represented by a **State Action Tree**. As one goes deeper into the problem, the number of possible action sequences increases rapidly.
  - **Seven General Problem Solving Techniques for searching the State Action Tree**:
    1. **Inference**: Deducing from explicit goals, givens, operations.
    2. **Classification of action sequences**: Organizing equivalent sequences into equivalence classes.
    3. **State evaluation and hill climbing**: Defining a quantitative evaluation function and choosing the action that moves closest to the goal.
    4. **Subgoals**: Breaking down the problem into simpler subgoals.
    5. **Contradiction**: Deriving an inference inconsistent with the goal state to eliminate possibilities.

6. **Working backward**: Beginning from the goal state and working backward.
7. **Finding relations between problems**: Identifying analogies with previously solved problems.

- **3.3 Newell's Approach**
    - Views problem solving as the **exploration of different paths to a solution** (like navigating a maze).
    - Objective problem structure is characterized by a set of states (initial, intermediate) and operators (moves).
    - **3.3.1 Summary of the Problem Space Hypothesis**:
        - For any problem, there are many alternative paths from initial to goal state, forming a **basic problem space**.
        - Human problem-solving is the production of knowledge states by applying mental operators.
        - Mental operators encode legal moves with restrictions.
        - People use knowledge and heuristics (like means-end analysis) to search the problem space.
        - These processes operate within the limits of cognitive system (e.g., working memory limitations).
        - The goal is to select one of the possible actions.
- **3.4 Problem Solving as Modelling**
    - A newer view: problem solving is the **construction of a situation-specific model or case model**.
    - The goal corresponds to a desired state of knowledge about the world.
    - The **case model** summarizes the person's understanding, leading to goal achievement.
    - Actions interact with the world to obtain knowledge. Problem solving builds models (domain model, task model).
    - **Competency theory**: Ties domain and task models together, defining what accomplishing the task means. It also describes rationality (e.g., reducing the problem's differential).
    - Problem solving is viewed as the **'creation' of a suitable case model**; interaction with the world is a resource for this.

Unit 4: Impediments to Problem Solving

- **4.0 Introduction**
    - Problem-solving attempts can fail if the problem space is not adequately searched or not well-represented.
    - **Einstellung** (mental set) and **functional fixedness** are classic obstacles to proper representation and search.

- **4.2 Effective Problem Solving**
    - Involves posing reflective questions about consequences and benefits of solutions (Haynes' model).
    - Requires considering basic issues, principles at risk, who benefits, and whether benefits outweigh harm (Hall's model).
    - Anticipating and adaptively responding to constraints, not viewing them as major impediments.
    - **Typical Processes**: Detailed thinking, explicit planning, collecting relevant information, monitoring progress, consulting others, planning follow-up.
- **4.3 Other Methods for Problem Solving**
    - **4.3.1 Einstellung (Mental Set)**: The tendency to **adopt a routine approach to problem solving** based on past experience, even when simpler solutions exist.
        - **Luchins' Water Jar Problems**: Demonstrated how people, after solving initial problems with a complex method (B-A-2C), would automatically try this method for later problems, overlooking simpler solutions (A-C), due to the established "set".
        - It is a form of **mindlessness** where people sample just enough features to confirm their set approach, rather than exploring alternatives. Constrains both problem representation and search.
    - **4.3.2 Understanding the Problem**: Requires clearer interpretation, devoting more effort to ill-structured problems, and seeing the immediate problem in a broader organizational context. In groups, adopt a broader range of goals with less personal stake in preconceived solutions.
- **4.5 Impediments in Problem Solving (General Reasons for Failure)**
    - Not being methodical.
    - Lack of commitment.
    - Misinterpreting the problem.
    - Lack of knowledge of techniques/processes.
    - Inability to use techniques effectively.
    - Using inappropriate methods.
    - Insufficient or inaccurate information.
    - Inability to combine analytical and creative thinking.
    - Failure to ensure effective implementation.
- **4.6 Productive Problem Solving and Impediments Thereof**
    - **Requirements**: Clear problem description (including negative factors, ownership, scope, consequences if unsolved), and a list of brainstormed solutions analyzed for reality, benefits, and consequences.
    - **Brainstorming rules**: Express all ideas, no idea too wild, quantity is important, group brainstorming is desirable, no criticism during brainstorming.
    - **Impediments to Productive Problem Solving**:

        - "Yes, but" attitude, intellectual defensiveness, fear (of incompetence, unaccepted ideas, being wrong).
        - Inability to be objective, not tuning into "inner child".
        - Emotional blocks: chronically immersed, discounting feelings, resentment, blaming others, believing problems are others' concerns.
        - Physical/mental fatigue, burnout, stress, anger, self-pity, depression.
        - Denial, bargaining.
- **4.7 Problem Solving in Teams and Small Groups (Barriers)**
    - Lack of structure, poor communication (misunderstanding, inattentiveness, dominance), no communication.
    - Negative attitude (inflexibility, closed-mindedness).
    - No goal or agenda.
    - Lack of participation from members.
    - Other problems: emotions, learning styles, gender, cognitive barriers, lack of transfer of structure between problems.
- **4.8 Critical Thinking in Problem Solving and Impediments**
    - **Three steps**: Becoming aware of assumptions, making them explicit, assessing their accuracy.
    - **Misconceptions about Critical Thinking**: It's wholly negative, leads to relativistic freeze, involves traumatic change.
    - **Other Fallacies**: Irrelevant reason, ad hominem attacks, false generalization, slippery slope, either/or, appeal to emotion.
- **4.9 Other Barriers to Problem Solving**
    - **4.9.1 Perceptual Blocks**: Inability to clearly perceive a problem or necessary information.
        - **Seeing only what you expect**: Obscures true nature.
        - **Stereotyping**: Not looking beyond the obvious.
        - **Delays in problem recognition**: Problems unnoticed until severe.
        - **Not seeing problem in perspective**: Too narrow a view, only part recognized.
        - **Mistaking cause and effect**: Leads to misdirected solutions.
    - **4.9.2 Emotional Blocks**: Perceiving a threat to emotional needs (achievement, recognition, etc.).
        - **Fear of making mistakes or looking foolish**: Most significant, difficult to overcome.
        - **Impatience**: Grabbing first solution, evaluating too fast.
        - **Avoiding anxiety**: Leads to risk avoidance, indecision, reliance on others, avoiding unfamiliar situations.
        - **Need for order**: Inability to cope with ambiguity/frustration.
        - **Lack of challenge**: If problem is routine, results in taking easiest route.

- **4.9.3 Intellectual Blocks**: Lack of necessary thinking skills or inability to use them.
  - **Lack of knowledge/skill in problem solving process**: Inadequate analytical/creative thinking, inflexible strategy, inability to use techniques.
  - **Lack of creative thinking**: Due to dominance of analytical thinking, lack of practice.
  - **Inflexible thinking**: Difficulty switching between thinking skills.
  - **Not being methodical**: Essential for effective solutions.
  - **Lack of knowledge/skill in using the 'Language' of the problem**: Difficulty with specialist jargon, inappropriate language use.
  - **Using inadequate information**: Leads to wrong conclusions.
- **4.9.4 Expressive Blocks**: Inability to communicate ideas effectively.
  - **Using the wrong language**: Using words for quantitative analysis.
  - **Unfamiliarity with a particular application of a language**: Difficulty with public speaking vs. writing.
  - **Inadequate explanations**: Lack of info or assuming audience knowledge.
  - **Passive/Dominant management style**: Prevents effective communication or makes others hostile.
- **4.9.5 Environmental Blocks**: External obstacles in social or physical environment.
  - **Management style**: Negative feedback leads to giving up.
  - **Distractions**: Noise, interruptions.
  - **Physical discomfort**: Creates distraction, stress, lethargy.
  - **Lack of support**: Missing information, advice, skills, resources, or encouragement.
  - **Stress**: Hinders creative thinking.
  - **Lack of communication**: Inability to get info, lack of encouragement.
  - **Monotonous work**: Dulls enthusiasm, leads to "automatic pilot".
  - **Expectations of others**: Influence performance/objectives.
- **4.9.6 Cultural Blocks**: Conditioning to accept what is 'expected' or 'normal'.
  - **Conformity**: Tendency to conform to established ideas/methods.
  - **Dislike of change**: Tradition preferred, need for security.
  - **Fantasy and humour are not productive**: Widespread belief, but innovators use them (unlikely combinations of ideas).
  - **Feelings, intuition, subjective judgements are unreliable**: Bias towards reason/logic, but intuition is key.
  - **Over-emphasis on competition or cooperation**: Unwillingness to listen to competitors, or avoiding new ideas to fit in.
  - **Taboos**: Actions/ideas excluded due to being distasteful, harmful, or contravening moral codes.
- **Other Person-Related Barriers**:
  - **Learning styles**: Visual, aural, step-by-step, introspective, discussive.
  - **Gender**: Some thinking styles gender-linked (females organize data more efficiently).

        - **Cognitive barriers**: Inability to identify and use concepts in analogous situations.
        - **Lack of transfer of structure**: Difficulty recognizing analogies between problems.
        - **Functional Fixedness**: Perception that an object or concept has only one use.
        - **Superficial transference**: Linking words/variables instead of deeper structures.
- **4.11 Functional Fixedness**
    - Discovered by Duncker (1935), it is the tendency to **see objects as having only a single, typical use**. Prototypical function dominates perception.
    - **Duncker's Candle Problem**: Participants were given candles, tacks, matches in boxes and instructed to mount candles on a wall. Only 43% solved when boxes were seen as containers. When boxes were emptied, all solved by using the boxes as platforms, demonstrating how breaking the "fixedness" helped.
    - **Coin Problem (Simmel, 1953)**: Finding a lighter counterfeit coin among eight using a balance twice. Initial fixation on dividing into two groups of four (due to 4+4=8 being highly available knowledge) hinders solution. The correct solution (dividing into 3+3+2) is unfamiliar and less obvious but allows finding the coin in two weighings. Overcoming fixedness often requires realizing obvious methods don't work, then reorganizing the problem.
- **4.12 Using Incomplete or Incorrect Representations**
    - Initial misinterpretation or focusing on wrong information hinders problem solving.
    - **Perkins' "Man at home" problem**: People assume "home" means a house, but the answer is a baseball game ("man coming a home"). This demonstrates how assumptions (a kind of mental set) and incorrect representations hinder solutions.
    - The **choice of representation makes a significant difference**: Using charts/flow charts is more successful than just writing facts.
    - **Numbers game**: A set of numbers (1-9) where players choose three to sum to 15, is much harder when viewed as numbers but easy when represented as a tic-tac-toe grid.
- **4.13 Lack of Problem Specific Knowledge or Expertise**
    - Applies to problems beyond simple puzzles, such as engineering, medical diagnoses.
    - **Experts and novices approach problems differently**. Familiarity with a domain changes problem-solving methods.
    - **Chess Study (de Groot, 1965; Chase & Simon, 1973)**: Chess masters and weaker players consider similar numbers of possibilities, but masters choose the best move more easily. Masters extract more information from brief exposures to game-configured chess boards (not random ones) than novices.
    - **Differences**:

        - **Representation**: Experts represent problems at a deeper, principled level (e.g., physics principles), while novices focus superficially (e.g., objects).
        - **Analysis**: Experts spend more time qualitatively analyzing/understanding the problem, while novices tend to jump into solutions.
        - **Error Checking**: Experts are more likely to check for errors in their thinking throughout the process.
    - Expertise alone is not always sufficient; other factors like medical and mental status can also affect problem-solving abilities.
- **4.10 Teaching and Learning Strategies that Enhance Problem-Solving Skills**
    - **Pedagogical Strategies (Teacher-centered)**:
        - Address emotional, psychological, and cognitive barriers simultaneously (e.g., open discussion to create a comfortable learning environment).
        - Promote **active involvement** (e.g., student learning groups) and experimentation.
        - Accept multiple solution attempts and encourage recording/discussing problem-solving methods.
        - Employ varied activities and approaches (e.g., graphics, practice, different interpretations of data) to cater to diverse learning styles.
    - **IDEAL Steps** (a common solution strategy):
        - **I**dentify the problem.
        - **D**efine and represent the problem.
        - **E**xplore possible solution strategies.
        - **A**ct on the strategies.
        - **L**ook back and evaluate.
        - Beneficial across many disciplines.
    - **Promoting Transfer**: Help persons identify common themes and recognize for productive problem-solving.
